

Introduction to Proof in UK Universities

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Abstract

There is a considerable rift between the mathematics studied by high school students and their counterparts studying mathematics at a university level. Whilst a component of this is the change in difficulty of the content studied, an arguably more important aspect is the change in the emphasis placed on proof. It is natural then that many universities run an *introduction to proof* course to try to bridge this gap. Whilst the content covered in many mathematics courses is somewhat standard across different institutions, the material covered in introduction to proof courses is extremely varied across UK universities, creating a wide variety in the attitudes that students have towards proof. As such, this project categorises the introduction to proof courses at Russell Group universities based on their content by quantitatively analysing their course syllabi. Moreover, in this project students are surveyed across these universities in order to conjecture the impact that the content to which students are exposed to during their introduction to proof has on their attitude towards proof.

Declaration

We declare that this project was composed by ourselves and that the work contained therein is our own, except where explicitly stated otherwise in the text.

(Alexander Bell, Pablo Ortuno Floria, Andrew Shaw, Chenrui Xu)

*This project is dedicated to Iain, the Kings Buildings' night
guard*

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Chapter 1

Introduction

Undergraduate mathematics degree programmes in UK universities have many tasks to accomplish in their first year. They are required to prepare students for more technical and rigorous mathematics in coming years, familiarise students with key topics, and also introduce a keystone in mathematics which many students may not have seen prior: proof. Given the largely computational style of mathematics taught in high schools throughout the UK and further afield [42, 54, 66, 67], the notion of proof can be challenging for students first encountering it [55]. As such, many universities offer an introduction to proof (ITP) course in the early stages of their mathematics degrees to give students some experience with this fundamental part of mathematics.

Much of mathematics is steeped in proof, and as such, the choice of content which could be utilised to introduce it is vast. This leads to a lack of standardisation amongst the content studied in ITP courses, a stark contrast to other introductory mathematics modules. David and Zazkis [12] address this variability by categorising 176 ITP courses in high and very high research output universities in the U.S. based on the content they use. One of the aims of this report is to categorise the ITP courses in the Russell Group in a similar way. This gives the opportunity to corroborate David and Zazkis' findings on an international scale, and provide groundwork for other avenues of research which these works give rise to. These may include, but are not limited to, the impact of the content studied in ITP courses on students ability to prove mathematical statements, whether universities should offer an ITP course as part of their undergraduate mathematics programmes, and ways in which these courses could be delivered to improve both the attitude and aptitude of students undertaking them.

The lack of standardisation across ITP courses may cause students to have very different experiences when being introduced to proof at university. As such, another aim of our project is to examine the attitudes towards proof which students have throughout their transition to tertiary mathematics education. At this point, it is important to note that student attitudes toward proof may be impacted by multiple factors, such as their experience of mathematics prior to university, the teaching style of ITP course lecturers, and the influence of their peers. For the purpose of this report, we acknowledge this limited scope, and as such make no comment on a defined relationship between the content covered and the student attitudes, but simply theorise the possible impact the content covered could have.

Chapter 2

Literature review

Since the incorporation of teaching and learning theories from psychology and sociology into the field of mathematical education during the 20th century¹ [65], mathematical education literature has been greatly preoccupied with the epistemological questions, applications, and pedagogical strategies which arise from such theories of learning, particularly in primary and secondary educational contexts. Nevertheless, research regarding tertiary education has been gaining popularity in recent years [31]. Within this field, many researchers have been particularly interested in the study of mathematical proof at this level [5, 32, 56], as it is perceived to be at the core of tertiary mathematical education [55].

It is widely agreed that students are not exposed to proof in a significant way during their secondary-school education [68], which has an impact on their conceptual understanding of proof. For example, CadwalladerOlsker [7] discusses how the development of a student's 'proof scheme'² could be affected by this lack of exposure. It is typical that earlier in their education, a high-school student may possess an *external conviction* proof scheme, by which they are convinced of the validity of a proof based on external factors (e.g. a teacher told them that a proof is true, so they believe it without question). When this same student embarks on a mathematics degree, they are suddenly expected to possess an *empirical* proof scheme, by which they are convinced of the validity of a proof because it 'makes sense' to them (for example, by means of previous examples or an alternative mental image of the concept that is being proved). Perhaps ideally, they would possess a *deductive* proof scheme, by which they are convinced of the validity of a proof because they are able to validate theorems based on logical deductions that hold for all cases³. This jump in understanding can be overwhelming for students, so it is unsurprising that many struggle when being introduced to proof in their mathematics degree.

Additionally, during students' transition from high-school mathematics to university mathematics, difficulties arise from the marked difference in language used [19, 55], the lack of preparation at high-school level to write proof [38, 72], and the dif-

¹i.e. the ideas of Skinner, Thorndike, Piaget, Vygotsky, Skemp, etc. [76].

²The *proof scheme* concept was originally coined by Harel and Sowder [28].

³These categories of proof schemes are divided further into more specific subcategories [27, 28]. We have opted not to discuss them as they fall outside of the scope of this project.

ferent epistemological perceptions of the mathematical world between academics and pupils [55]. Although these difficulties are largely reported from findings within the US education system, it is necessary to note for the context of this report that these problems are also prevalent for students in the UK [22].

These difficulties during the transition from secondary to tertiary mathematics education become particularly apparent when considering student aptitudes towards proof. The widespread summative assessment diets used in secondary schools during the second half of the twentieth century, which measured student's performance through standardized multiple-choice testing [16], provided a common ground on which to study pupils' aptitudes for mathematics. This type of assessment emphasised aptitudes in computation, simple problem solving, and geometry, and did not test proficiency on more sophisticated tasks, thus measuring mathematical ability in the same way as a simple skill because this was convenient logistically [44]. Nevertheless, changes in academic and institutional views of assessment [16] have led to the consideration of mathematical aptitude as something more complex than just "being able to carry out computations". This paradigm shift is at the core of some research regarding the proving process and so different aspects of it have been studied in recent years. Firstly, meta-mathematical abilities which aid the learner in their cognitive processes such as an internal monitor, strategic knowledge, and concept usage [71]. Secondly, proof validation [57] and the role which language plays in the process of reading, understanding and assessing a proof [39]. Thirdly, proof construction, which has been studied from a cognitive lens [1], and also from a linguistic one [39], in particular, the peculiarities of the *mathematical register*⁴ [3, 45].

With this further consideration of the 'dimensionality' of the mathematics learner many researchers have opted to study students' perception and attitude towards proof [55]. For example, Furinghetti, Matos, and Menghini [20] suggest that students who perceive mathematics at a tertiary level to be a continuation of secondary level mathematics and realise that the changes of notation and formalisation through proof are a necessity will prevail during the secondary-tertiary transition. De Villiers [70] suggests that students' perception of proof is often narrowed due to their previous experience with the subject, and then goes on to provide five separate 'functions of proof' as a basis for future research on student perception of proof. Bleiler-Baxter and Pair [6] expand on this concept by exploring how students' 'proving activities' relate to these functions of proof.

Researchers have come to various conclusions about the links between attitude and aptitude for proof: students with high aptitude for proof tend to have a better attitude towards proof [61]; students with a good self-efficacy towards proof (i.e. a good attitude towards proof) tend to have a higher aptitude for proof [20, 29]. However, there is some disagreement regarding these conclusions. For example, Moore [38] suggests that attitude affects the type of proof that students produce rather than the quality of said proof. Having said that, it is the case that certain types of proof may be considered to be of a lower quality than others, in particular where the proof's constructor (writer)

⁴This corresponds to the use of mathematical discourse using some natural language.

and the proof's validator (reader) have different attitudes with regards to communication and rigour. This relates to the different views on mathematical proof which have arisen and, in particular, the clash between formalist and subjective schools of thought⁵ [7]. Regardless, we can reasonably assume that attitude towards proof, aptitude for proof, and the types of proof that a student may produce are all related.

While proof is a widely researched subject, it is largely referred to as a content-general (or content-independent) mathematical activity; Students are expected to be able to prove in a wide variety of mathematical contexts, independent of the content to which they may have been exposed to whilst first being introduced to proof. Dawkins and Karunaran [14] suggest 3 reasons as to why this may be the case:

1. **Psychological reasons.** Experts generalise the experience of proving across topics because they have a high aptitude for it, whereas novices may not.
2. **Empirical findings.** There is much evidence across different topics of students' difficulty dealing with proof, so it seems natural to consider proof as an activity that is separated from the content it pertains to.
3. **Analytic process.** Educators have only been interested in proof as a broad concept in itself.

Indeed, Dawkins and Karunaran [14] conclude that a content-general approach is an adequate way to *describe* recurrent phenomena of student proving, but one must consider the mathematical content at hand to *explain* why these phenomena occur. As such, they highlight the importance of content-specific analysis when studying and researching proof in an educational context. David and Zazkis [12] developed this by creating groundwork for further content-specific analysis of proof for populations of students in US universities by categorising introduction to proof (ITP) courses based on the content contained in their syllabi. In fact, Dawkins and Karunaran [14] are not alone in suggesting that content should be more seriously considered when researching ITP courses⁶. Alcock and K.Weber [1] conclude that content often dictates which approach, *referential* (using concrete examples to understand and solve problems) or *syntactic* (using formal structures and logic without relying on examples), might be more suitable or effective in proving a concept. Similarly, Baker and Campbell [4] researched the recurring problems students encounter when writing mathematics in ITP courses. Hanusch [25], and Alcock and K.Weber [1] place importance on the types of examples that are used in the teaching of proof, and suggest that more research should be done to determine whether certain topics are of more use when constructing these examples. Likewise, Selden and Selden [57] highlight the impact that the content a student is exposed to has on proof validation and production.

Nevertheless, the research presented that addresses the need for content-specific analysis in proof seems to mostly be in the context of the study of students' *aptitude* for proof in the US education system. We therefore identify two gaps in the current state of mathematical education literature: Firstly, there is a lack of content-specific

⁵A detailed explanation of formalist and subjective views can be found in [subsection 3.2.5](#).

⁶Savic [53] provides a summary of this view, criticising the current state of the field.

research around proof within UK universities, and secondly, there is a lack of research which explores the relation between content exposure and student *attitudes* towards proof. This project seeks to contribute to these well-defined gaps by trying to answer the following two research questions:

RQ1: How can ITP courses in UK universities be categorised based on their content?

RQ2: How does the content to which students are exposed to in an ITP course affect their attitudes towards mathematical proof?

Chapter 3

Methods

To answer **RQ1**, we constructed a categorisation of ITP courses in UK universities based on their content using quantitative methods, which we describe in section **3.1**. In section **3.2**, we detail how we surveyed and analysed attitudes of undergraduate mathematics students in order to answer **RQ2**.

3.1 Categorising ITP courses based on content

In this section, we describe the methodology used to categorise the ITP courses in the Russell Group¹. In subsection **3.1.1** we describe the selection criteria for these courses, the reasons for restricting the scope of our research to Russell Group universities, and the process of collecting data from each courses' syllabus. This data was then interpreted and processed, procedures described in subsections **3.1.2** and **3.1.3** respectively. In subsection **3.1.4**, our categorisation of ITP courses based on their content using a quantitative approach is described.

3.1.1 ITP criteria and syllabi collection

To ensure that a given ITP course was the first introduction that students had to proof in their tertiary education, we restricted our attention to first and second year courses in UK universities. In the case that two courses qualified to be categorised as ITP, one in first year and one in second year, the course prescribed for first year was chosen. This ensured that the chosen course was the true introductory one, and that student experiences would not be biased by former courses. It was also a requirement that this course was a mandatory part of the mathematics undergraduate degree programme.

We decided that the Russell Group provided a distinct subset for this study. Further, all of the Russell Group universities offer an undergraduate mathematics degree, a clear necessity for this project. We searched this degree programme at each university for the appearance of an ITP course. A maximum of one course was to be chosen to be an ITP course at each university. So, when we encountered multiple courses at a single university which appeared to meet our criteria, we conducted a review of their

¹The Russell Group is an association of 24 British universities (out of the total of 166) characterised by an intense research focus and a high quality of teaching [52].

syllabi to determine which course placed a greater emphasis on proof and mathematical rigour. This and the above paragraph provides the definition of an ITP course used henceforth in this report: the first course in a university mathematics undergraduate programme which *specifically* aims to introduce students to proof, and as such places a high emphasis on reading, constructing and communicating proof.

Elaborating further, we studied each university's webpage individually for publicly available information regarding their courses. When we found a course which was believed to satisfy our requirements, its syllabus was uploaded to a Microsoft Teams folder to be inspected by other group members individually. Then, we collectively decided whether each university's mathematics degree programme contained an ITP course, and if so, which course to label as such. Data was also collected surrounding the logistics of each ITP course, with the aim of checking any trends or similarities amongst the Russell Group. The logistical aspects of the courses that were captured and their descriptions are summarised in Table 3.1.

<i>Logistical data</i>	<i>Description</i>
Location	Geographic location of the university at which the ITP course in question is taught.
Course code	Code associated with each ITP course.
Textbook and authorship used	The textbook and authorship on which the course is based, if applicable.
Reading list	Recommended texts for students studying the course. They do not comprise essential material, but have been deemed to be useful/relevant by the course organiser.
Syllabus date	Year on which the course has been delivered.
Delivery year	Year of the degree on which the course is delivered.
Prerequisites	Courses in or outwith the university which students are required to have studied prior to taking the ITP course in question.
Credits allocated	Number of credits which the module takes up.
Assessment information	Information regarding how the ITP course is assessed.
Learning and teaching activities	Number of hours allocated to workshops, lectures, independent study, etc.

Table 3.1: Logistical data taken from university webpages.

It should be noted that one university had minimal information on their courses on their website, and so an external website, StuDoc [59], was used to obtain the lecture notes of this course². Therefore, the reliability of information regarding the content described for this university is lesser when compared to official university documents. This experience underscores the challenge of accessing up-to-date and comprehensive course details directly from university websites. Such difficulties highlight a broader issue within the academic community: the pressing need for universities to enhance the transparency and accessibility of their course information. Making such data more readily available would not only facilitate academic research but also support the development of innovative educational programmes.

3.1.2 Content categories

During our examination of the syllabi, we identified twelve major topics used to teach proof: Real analysis; calculus; complex numbers; discrete mathematics; functions and relations; group theory; linear algebra; logic; number systems; number theory; proof techniques; and set theory. We refer to these as *categories* of content to avoid confusion when referring to the more granular topics used in such courses, which we denote as *topics*. For example, "convergence" is a topic in the category "Real analysis". In 7 universities, we encountered topics which did not fit into any of these categories, which motivated the inclusion of an "Other" category. These varied from *simple axiomatic probability*, to *writing proofs in the programming language LEAN*. Table 3.2 provides a summary of the most common topics found in each content category (Appendix A).

3.1.3 Quantitative measures

The content data was then transferred to a separate excel sheet. This involved collating certain topics together to avoid our insight becoming overly granular. For example, individual topics such as "injective functions" and "surjective functions" were combined into "injective and surjective functions". This gave rise to a set of common topics in each category, which helped assess the usage of each content area within a given ITP course. In each category, we included a topic named "present but no detail", which we used to indicate the case when the category was mentioned in a course's syllabus, but no information on the topics covered was detailed.

We calculated multiple statistics to give our categorisation quantitative validity. Firstly, we defined the *weight* of an individual category in an ITP course as follows:

$$\text{Category weight} = \frac{\text{Number of topics covered in the category}}{\text{Number of topics covered in the syllabus}}. \quad (3.1)$$

This number, represented as a percentage, allowed us to see how much each content category contributed to a course's syllabus. While this does not take the significance of different topics into account (for example, within Complex Numbers, *complex conjugates*, although essential, is a much less significant topic than *the Fundamental The-*

²The official syllabus was found at a later date and all topics listed in the document were consistent with the ones that we had identified from the lecture notes.

orem of Algebra), it still acts as an appropriate metric to approximate the amount of emphasis placed on a category, given that if more topics are covered, then it is likely that more time has been spent on that category. Regardless, it is still important to consider this as a limitation. When a category was marked as “present but no detail”, we deemed the number of topics covered in the category to be: the average number of topics covered in the category across all of the courses in which the category was present. Secondly, for each course, we calculated the *standard deviation* of category weights³. Smaller standard deviations would indicate that the content categories covered in a course were given a similar emphasis, while higher standard deviations would indicate that some categories were given significantly greater emphasis than others. Thirdly, a course’s *variability* was defined as:

$$\text{Variability} = \frac{\text{Number of categories covered}}{\text{Total number of categories}}. \quad (3.2)$$

This number, represented as a percentage, allowed for an easier visualization of how many content categories were covered in a particular course.

<i>Content category</i>	<i>Topics</i>
Real analysis	Sequences, convergence, limits, series.
Calculus	Standard derivatives, derivatives by first principles, integrals, differential equations.
Complex numbers	Basic properties, roots of unity, polar form, de Moivre’s theorem.
Discrete	Binomial theorem, permutations, countability.
Functions and relations	Functions, injections, surjections, bijections, equivalence relations.
Group theory	Examples of groups, basic properties of groups, subgroups, group homomorphisms.
Linear algebra	Vectors, vector spaces, systems of linear equations, lines and planes in 3-dimensional euclidean space, matrices.
Logic	Quantifiers, logical operators, implication, negation, converse, contrapositive.
Number systems	Real, rational, and irrational numbers.
Number theory	Prime numbers, prime factorisation, modular arithmetic, congruence.
Proof techniques	Induction, contradiction, direct proof, general proof comprehension.
Set theory	Countability, notation, basic set theory.
Other	Polynomials, writing proofs in LEAN, communicating mathematics.

Table 3.2: List of the most common topics in each content category.

³Throughout this report, whenever we mention standard deviation, it is in reference to the standard deviation of category weights.

3.1.4 Categorising and subcategorising ITP courses

Using the setup detailed in subsections 3.1.1, 3.1.2 and 3.1.3, we were able to categorise the ITP courses based on their content by considering the following factors in descending order of importance: their two highest category weights and the difference between them; the standard deviation of their category weights; and the course variability.

The courses were grouped into three main categories, “Standard”, “Medley” or “Topic”⁴. These courses then had the opportunity to be further subcategorised by labels of the form “Category + Subcategory” in the case that there was significant emphasis on another category. Subcategories fell under the same Standard, Medley or Topic labels. This led to all ITP course being categorised as either “Standard/Medley/Topic only” or “Standard/Medley/Topic + Standard/Medley/Topic”.

We first deal with the overall categories before defining further subcategories. The appearance of four content categories in over 85% of the surveyed ITP courses (proof techniques 95%, set theory 95%, functions and relations 90%, and logic 85%) seen in table 3.3, suggests the existence of a standard set of content areas used to teach proof across the Russell Group, which is consistent with the one suggested by David and Zazkis [12]⁵.

<i>Category</i>	<i>Frequency</i>	<i>%</i>	<i>Average weight</i>
Proof techniques	19	95%	12.9%
Set theory	19	95%	9.0%
Functions and relations	18	90%	14.5%
Logic	17	85%	15.1%
Real analysis	14	70%	18.4%
Number systems	14	70%	12.9%
Number theory	12	60%	23.0%
Complex numbers	9	45%	14.8%
Other	7	35%	4.6%
Discrete	7	35%	7.0%
Group theory	6	30%	12.7%
Linear algebra	5	25%	45%
Calculus	3	15%	18.2%

Table 3.3: Frequency and average weight of each content category, and the percentage of courses that they appear in.

Henceforth, we refer to proof techniques, set theory, functions and relations, and logic all together as *standard content*. A course was categorised with the label “Standard” if

⁴These categories are described and discussed in sections 4.1 and 5.1 respectively.

⁵David and Zazkis [12] define standard content as the following set of topics: “symbolic/formal logic, truth tables, propositions, quantifiers, methods of proof (including contradiction and induction), number systems, sets relations and functions, infinite sets, and cardinality”.

there was a clear emphasis on this standard content, whilst displaying a low variability and a medium standard deviation. A course was categorised with the label "Medley" if there was no clear emphasis on any specific topic (standard or otherwise), quantitatively indicated by high variability and a low standard deviation. Finally, a course was categorised with the label "Topic" if there was a clear emphasis on one (non-standard) topic, shown quantitatively by a low variability and a high standard deviation.

Amongst the Standard courses, a course gained a "+Medley" label if their non-standard content had the characteristics of a Medley course. A course gained a "+Topic" label if its non-standard content had the characteristics of a Topic course. Finally, a course was categorised as "Standard only" if a very clear majority of the content was standard (+70%), and the non-standard content did not display Medley or Topic characteristics.

Amongst the Medley courses, a course gained a "+Standard" label if there was an emphasis on standard content (with all four of the standard content categories present) that was outweighed by the remaining medley-style content. A Medley course gained a "+Topic" label if there was an emphasis on a particular topic amongst the medley-style content. It was crucial that this topic did not constitute the majority of the course. A course was categorised as "Medley only" if it was centred around a broad spread of content and there was no clear emphasis on a specific topic, be that standard content, or otherwise.

Amongst the Topic courses, a course gained a "+Standard" label if there was an emphasis on standard content that was outweighed by the main topic. A course gained a "+Medley" label if the remaining content was significantly weighted and varied, displaying the characteristics of a Medley course. Note here that the main topic should still outweigh the other medley-style content, otherwise, the course should be categorised as "Medley + Topic". A course gained a "+Topic" label if there was a secondary topic which also had a significant weight, but did not outweigh the main topic. A course was categorised as "Topic Only" if the vast majority of the course's content (+60%) was attributed to the topic and if the remaining content did not display clear characteristics of standard, medley or topic content.

The only other distinction to be made was the addition of the "Standard*" label. This denotes a Standard course which contains all standard content besides one content category, and in which there is still significant emphasis on standard-style content. For instance, a course containing large proportions of Set theory, Functions and Relations, and Proof techniques but with no appearance of Logic could be given the Standard* label. Courses missing more than one of the standard content categories were not classified as Standard*, since these two (or one in the case that three were missing) content categories were either given significant emphasis, showing characteristics of a Topic course, or minimal emphasis, showing characteristics of a Medley course.

We now exemplify the categorisation process by categorising the ITP course at the University of Edinburgh, *Proofs and Problem Solving*. Figure 3.1 shows the granular breakdown of the content categories used in the course and their weights. The first

pie chart displays a wide variety of content categories used. Further, when we group standard content in the second pie chart, we observe that:

1. Similar emphases are given to Standard content and to Real analysis content, making up for around a quarter of the course each.
2. Similar emphases are given to Number theory, Complex numbers, Discrete, and Number systems, making up for around a tenth of the course each.

Although this course gives a relevant emphasis to Real Analysis and Standard content, individually they do not constitute a significant enough portion of the course syllabus for this course to be categorised as Standard or Topic. Moreover, this course covers a wide variety of content areas, further exemplified by a high variability of 69%. Hence, we decided it should be categorised as Medley. For two more examples of the categorisation process, see Appendix C.

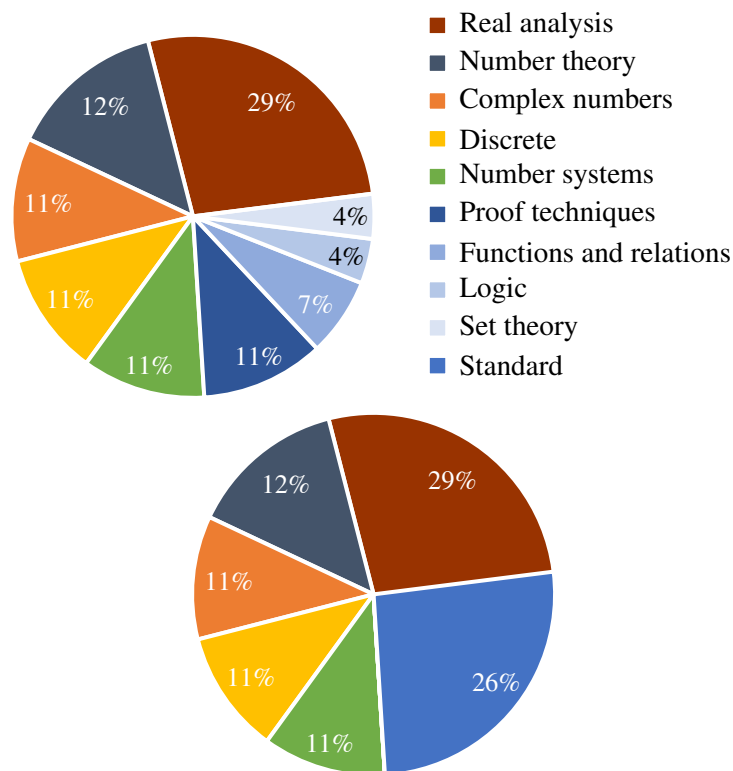


Figure 3.1: Category weights of the ITP course at the the University of Edinburgh first without, and then with standard content grouped.

3.2 Surveying student attitudes towards proof

In this section we describe the methodology that we used to investigate student attitudes towards proof, which were collected using a survey. We consider the aims of it in subsection 3.2.1, its construction in 3.2.2, and the rationale behind each question in 3.2.3. We then describe the survey's distribution in 3.2.4, and conclude in subsection 3.2.5 with the methodology used to process the gathered data and produce the results seen in section 4.2.

3.2.1 Aims

One of the aims of the survey was to investigate how the category of ITP course that a student is exposed to affects their attitude toward proof. We initially considered also investigating how ITP category affects aptitude for proof, but decided that this was outside the scope of the project⁶.

We were also interested in the perceptions students had about their institution's ITP course. Therefore, a secondary aim of the survey emerged: to investigate whether students believed that they had taken a course that had prepared them to read and write mathematical proof, and whether this course was the same as the one which we deemed to be the ITP course for their university. This could help us determine whether our definition of an ITP course was accurate and valid. Responses could also shed light on whether a student believed that an ITP course had effectively introduced them to proof.

3.2.2 Preliminaries

We included an introductory section to the survey to ensure that participants consented to taking part and that we had suitable means to extract a participant's response should they want their answers to be removed. This was essential as the survey required participants to share personal data (university that they attended, course that they had taken, and their year of study). Our research was conducted under approval from the ethics officer at the School of Mathematics at our institution, to ensure that we complied with GDPR regulations.

To fulfil the first aim of the survey, it was necessary to consider not only populations of students who had recently been exposed to an ITP course, but also a population of students who had not yet been exposed to an ITP course to act as a control group. This would allow us to make inferences regarding student attitudes before and after undertaking an ITP course of a specific category. The second aim of the survey was not applicable to the control group since they had not yet been exposed to an ITP course.

⁶To effectively measure the impact which a course has on student aptitudes, we would have to create a series of proving tasks (and a means to measure the results of these) and consider a great number of factors for which we did not have the logistical capabilities, for example the teaching environment, or the ratio of students to lecturers/tutors. We would have to also consider pedagogical strategies, course format, and course materials.

Therefore, our survey consisted of the following sections:

1. Consent and screening,
2. Conceptual understanding,
3. Introduction to proof courses,

with the final section being omitted from the survey distributed to the control group.

We decided to implement our survey using Microsoft Forms for two reasons. Firstly, our institution's servers operate with the Microsoft 365 platform so we could gather and store all the data in secure digital locations through our institutional accounts, ensuring high data protection standards. Secondly, Microsoft Forms' built-in implementation of Microsoft Excel allowed us to export the data directly into an Excel spreadsheet, which was already our program of choice to work with the syllabi data (section 3.1).

3.2.3 Questions

Each survey question is detailed below, together with the reasoning behind it. We consider each section of the survey as detailed above (subsection 3.2.2) separately.

Consent and Screening

Participants were first presented with information about the purpose of the study, why they had been invited to take part in the survey, data protection and details about how their data would be anonymized. The first four questions functioned to gain a participant's consent and ensure that they were a member of the target population for the survey. These first four questions can be found in Appendix B

The remaining questions in this section function to characterise the participants:

5. *Which University do you attend?* **[choice from drop-down list]**

This question allowed us to categorise responses based on what category of ITP course they took⁷.

6. *What is your highest level of qualification?* **[choice from drop-down list⁸]**

We used this question to explore whether a students' previous education has any impact on their attitude towards proof.

⁷This question was not asked to the control group, given that this group came solely from our institution, as discussed in 3.2.4.

⁸This drop-down list contains 'SQA Highers', 'SQA Advanced Highers', 'A-Levels', 'International Baccalaureate' and an open text box for 'other' qualifications which may have appeared in smaller quantities.

Perception of proof

This section addresses the questions regarding student attitudes towards the concept of proof.

7. *Please indicate whether you agree or disagree with the following statements.*
[5-point Likert scale⁹]

- *I feel comfortable reading and understanding mathematical proof.*
- *I feel comfortable using techniques that I have learned to construct mathematical proof.*
- *I enjoy communicating ideas using mathematical proof.*

This question allowed us to directly compare different cohorts' attitudes towards understanding, constructing and communicating proof.

8. *What is a mathematical proof to you?* **[open text response]**

This question allowed us to explore explicit attitudes towards proof, the view that students have on proof, and their perception of the role of proof.

Introduction to proof courses

This section addresses questions concerned with student attitudes towards proof-based courses at their own institutions. These questions were only asked to students who had already had an opportunity to be exposed to an ITP course.

9. *Is there a module that you feel particularly aimed to prepare you to read and write mathematical proof?* **[yes/no]**

If participants answered yes, they were shown the remainder of the questions in this section. If they answered no, then the rest of the questions remained hidden.

10. *What is the name of this module?* **[open text response]**

Along with question , this question allowed us to identify whether a student believed that the course that we deemed to be the ITP course at their university aimed to introduce them to proof.

11. *Please indicate whether you agree or disagree with the following statements.*
[5-point Likert scale]

- *This module succeeded in preparing me to read and write mathematical proof.*
- *I enjoyed this module.*

These statements allowed us to analyse different cohorts' attitudes towards proof-based courses at their institution.

⁹A 5-point Likert scale requires participants to indicate whether they 1. strongly disagree, 2. somewhat disagree, 3. neither agree nor disagree (abbreviated as *Nand*), 4. somewhat agree or 5. strongly agree to each statement provided.

12. *Is there anything that you would change about the mathematical content of this module?* **[open text response]**

This question allowed us to gain insight into the content that students believed would be taught in an effective ITP course.

3.2.4 Distribution

First-year students at our own institution acted as the control group¹⁰. A reason for this is that ITP courses at different institutions run during different semesters. The ITP courses we identified all ran during the first year of their respective mathematics degree programmes, so second-year students were guaranteed to have been exposed to an ITP course (in the case where their institution delivered one). Hence, we distributed our survey to first-year students in our own institution, and to second-year students at multiple universities.

At our own institution, the survey was distributed to eligible students by two research members. This involved four presentations before the students' lectures - two to the cohort of students in first year, prior to their first exposition to proof in the ITP course, and two to the cohort of students in second year, who had already been exposed to proof during the ITP course, but prior to any other proof course which could potentially impact their attitudes. These presentations consisted of a short, two-minute talk on the aims of the project, the reason for the student's eligibility for the survey, and the possible usefulness of the project for future students. It was stressed that this survey had no impact on the students' grades positively or negatively and that they were not obligated to complete it.

Other Russell Group universities were exposed to the second-year survey by choice, through the same contact email that was used to gather data on the ITP courses. Lecturers were asked whether they would like more information on the survey, and in the case that they answered yes, a brief summary of the aims of the survey and the project was given, as well as a QR code that they could distribute to students which would lead them to the survey. In the case that they declined, or did not address the question regarding the survey, no further contact was made. Our lack of insight into the distribution of the survey at other institutions motivates some limitations and potential biases in the analysis of these responses. Indeed, it is unclear to what extent students were advised or motivated to complete the survey.

3.2.5 Processing gathered data

Using the responses to the first four questions, we filtered out any responses that were invalid due to participants not consenting to us collecting their responses or not being a part of our target population.

¹⁰Note that the student attitudes pre-exposition to an ITP course cannot be assumed to be consistent across institutions. Indeed, previous education could have an impact on this. We note this limitation and make inferences on the comparison between those who have and those who have not been exposed to an ITP course accordingly.

<i>Likert Scale Response</i>	<i>Score</i>
Strongly agree	2
Somewhat agree	1
Neither agree nor disagree	0
Somewhat disagree	-1
Strongly disagree	-2

Table 3.4: Attitude score assigned to each Likert scale response.

Processing responses to Likert scale questions

We used the responses to question 7 to analyse positive and negative attitudes towards proof. For each response, we assigned three "attitude scores" (one for each statement) based on their choices on the Likert scale. Table 3.4 shows the assigned score to each Likert scale response. The scores for each statement were added up to produce a final "attitude score" for our analysis and we defined three ranges to group responses as *positive* (2 to 6), *neutral* (-1 to 1), and *negative* (-2 to -6). We justify our choice of such ranges by detailing the process of defining the positive range. When analysing the different combinations for a set attitude score, we found that the threshold of what we would consider a positive attitude was a score of 2. For example, a student could have responded "strongly agree", "strongly agree", and "strongly disagree" for this question (with scores of 2, 2 and -2, respectively). In this case, we would consider the presence of "strongly agree" twice as an indicator of an overall positive response despite the presence of a "strongly disagree". We examined the other combinations that gave an overall score of 2, as shown in Table 3.5. A similar justification process was used to define the neutral and negative ranges. For question 11 we interpreted the responses at face value and did not process them any further.

<i>Question 1</i>	<i>Question 2</i>	<i>Question 3</i>
Strongly agree (2)	Strongly agree (2)	Strongly disagree (-2)
Strongly agree (2)	Somewhat agree (1)	Somewhat disagree (-1)
Strongly agree (2)	NAND (0)	NAND (0)
Somewhat agree (1)	Somewhat agree (1)	NAND (0)

Table 3.5: Different response combinations which have an attitude score of 2. "Neither agree nor disagree" is abbreviated to NAND.

Encoding text-based responses

Text-based responses to question 8 were encoded deductively. We aimed to extract three pieces of information from each response. Firstly, explicitly negative and explicitly positive attitudes towards proof. Secondly, views towards proof. We used CadwalladerOlsker [7]'s characterisation of formalist and subjective views as our two labels to categorise respondents' views. Formalist views are more concerned with the

use of formal logical language in proof and its rules of inference to show that areas of mathematics are "free of contradictions". Subjective views are more concerned with the inherent communication aspect of proof: proofs are only accepted by the mathematical community if they can be read and understood. Lastly, we aimed to capture students' perception on the roles of proof. We used de Villiers [70]'s characterisation of these roles and decided to include "Intellectual challenge" due to CadwalladerOlsker [7]'s characterisation of it as a role of proof. In Table 3.6 we provide basic examples of codewords we use in our analysis. Moreover, we include three examples of how we coded text-based responses in Table 3.7.

<i>Attribute</i>	<i>Aspect</i>	<i>Codewords examples</i>
Attitude	Positive	Beautiful, pretty, elegant, nice
	Negative	Bad, difficulty, nightmare
View	Formalist	Logic, exact, axioms, formal, rules, truth
	Subjective	Argument, rigour ¹¹ , convincing, accepting
Role	Verification	Truth, demonstrate, (desired) conclusion
	Explanation	Why, explain, show
	Systematization	Theorems, definitions, axioms, collection
	Discovery	Research, discover, find
	Intellectual challenge	Encourage, enjoyment, motivation
	Communication	Illustrate, communicate, read

Table 3.6: Examples of codewords used in our analysis of open-text responses.

<i>Response</i>	<i>Attitude</i>	<i>View</i>	<i>Roles</i>
Showing using the tools of analysis that some proposition is undeniably true, or at least consistent with the first principles of mathematics	N/A	Formalist	Verification, explanation
A way of displaying how a result was found	N/A	Subjective	Communication, discovery
Nightmare	Negative	N/A	N/A

Table 3.7: Examples of our processing of text-based responses.

Chapter 4

Results

We identified ITP courses in 87.5% (21/24) of the Russell Group Universities. In section 4.1, we detail how the variation in the content used in these ITP courses prompted us to categorise them using three different labels: *Standard* (4.1.1), *Medley* (4.1.2), and *Topic* (4.1.3).

The limited number of responses to the survey from students outside of our own institution affected our aim of exploring connections between student attitudes towards proof and the content with which they are introduced to proof. Nevertheless, in section 4.2, we present results on:

1. Students who have not yet been exposed to an ITP course (4.2.1)
2. Students who were exposed to a Medley course (4.2.2)
3. Students who were not been exposed to an ITP course because their institution does not offer one (4.2.3)
4. Students who were exposed to a Standard course (4.2.4).

4.1 Results regarding the categorisation of ITP courses

Following the ITP course labels defined in section 3.1, we categorised ten courses under a "Standard" label, seven under a "Medley" label, and four under a "Topic" label, with a further subcategorisation of each course. In this section, we first present some general results regarding ITP courses in Russell Group universities, followed by the specific results for each ITP course label in subsections 4.1.1, 4.1.2, and 4.1.3.

ITP courses in Russell Group Universities seem to prefer using a varied amount of content, with a mean of 7.05 content categories per course (table 4.1). The average weight of a content category in an ITP course was 13.6%, with most of the identified categories making up between 0% and 20% of a course's syllabus. In contrast, there were some outliers accounting for between 60% and 70% of certain courses' syllabi (Figure 4.1).

<i>Statistic</i>	<i>Amount of categories</i>	<i>Variability</i>
Minimum	4.00	30.8%
First quarter	5.50	42.3%
Median	7.00	53.8%
Mean	7.05	54.2%
Third quarter	8.50	65.4%
Maximum	13.0	100%

Table 4.1: Summary statistics of the number of categories present and variability in ITP courses.

This distribution of the average content category weight along with the distribution of the variability metric (defined in section 3.1.3) seen in table 4.1, motivated us to deviate from using David and Zazkis’s [12] suggested "Sampler" category: a course category defined as “a course using three or more topics”¹. In fact, the minimum number of content categories used in any of the ITP courses was four, so this label was not suitable for our analysis.

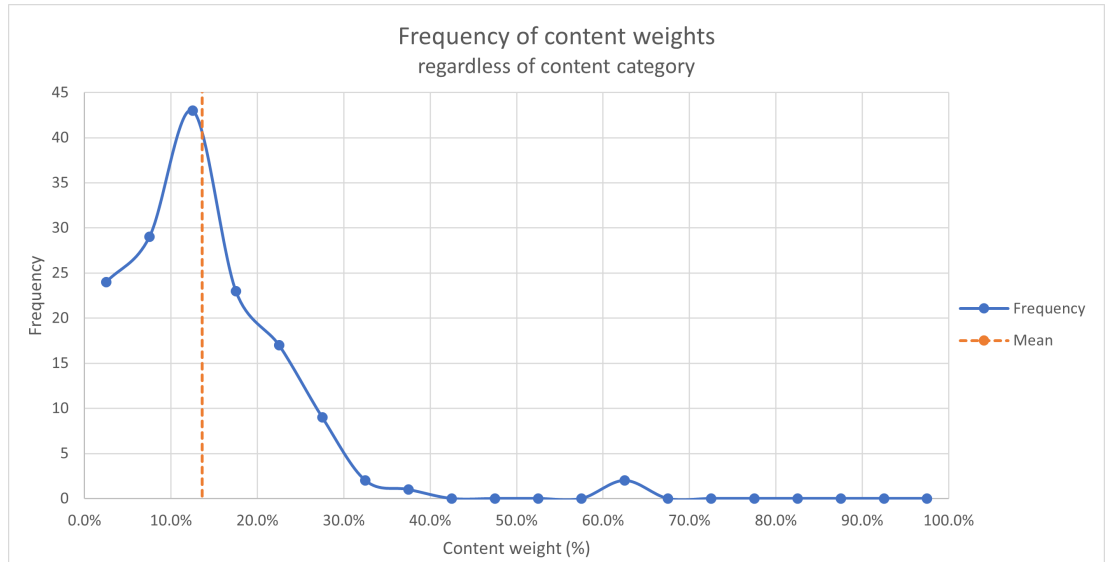


Figure 4.1: Frequency of content category weights, in 5% bands, across all surveyed ITP courses.

4.1.1 Standard courses

Standard courses account for 50% (10/20) of the identified ITP courses, and divided them into 3 subcategories: Standard Only (4/10), Standard + Topic (5/10), and Standard + Medley (1/10). The subcategorisation of Standard courses is summarised at the end of this subsection in table 4.3. Table 4.2 shows the frequency of each content category and its average weight across Standard courses.

¹David and Zazkis’ definition of *topics* is equivalent to our definition of *content categories*.

<i>Content category</i>	<i>Frequency</i>	<i>%</i>	<i>Average weight</i>
Proof Techniques	10	100%	14.9%
Set Theory	10	100%	12.2%
Functions and Relations	9	90%	20.8%
Logic	9	90%	19.9%
Real Analysis	6	60%	11.8%
Number Systems	6	60%	16.0%
Number Theory	4	40%	21.3%
Complex Numbers	3	30%	12.0%
Discrete	3	30%	8.00%
Other	2	20%	7.50%
Linear Algebra	1	10%	11.0%
Group Theory	0		
Calculus	0		

Table 4.2: Frequency and average weight of each content category, and the percentage of Standard courses that they appear in.

We explored the logistics of Standard courses (table 3.1). Standard courses ranged between a weighting of 10 and 20 credits in a university year of 120 credits, and all of them were single semester courses. We obtained assessment information from six of the ten Standard courses we found. Of these, five contained a coursework element which contributed to students' final marks, with the others simply using two open book exams to determine the students' grades. All Standard courses which detailed information regarding recommended student input hours (5/10) suggested 10 hours per credit, in line with guidelines from the Quality Assurance Agency for Higher Education [47]. Of the ten standard courses, there was information on the textbook and authorship used for one course, which used lecture notes written and compiled by the course lecturer. Further, considering the reading lists, two courses provided a set of texts recommended for students, with two and five texts in each respectively. There were no common texts between them.

Standard Only

Standard Only courses account for 40% (4/10) of the courses labeled as Standard. Over 70% of these courses' content consisted of standard topics, with small emphasis given to other topics to compliment the material². Of the Standard Only courses, one was labelled as "Standard* Only" since 80% of its content fell within standard content, but did not use any topics in Functions and Relations.

Standard + Topic

Standard + Topic courses account for 50% (5/10) of the standard-labeled ITP courses. One course was identified as being mostly standard (around 50% of its content), but

²Looking at course details, this complimentary material was either used to exemplify proof in other contexts, or acted as a brief introduction to future courses in the degree's curriculum.

without using any topics in Logic. We chose to include it in this category and identify it as "Standard* + Topic". We found the most common topic used to compliment standard content to be Number Theory (3/5). The two other courses labelled in this category used Real Analysis and Number Systems respectively.

Standard + Medley

Only one course was identified in this category, accounting for 10% of the Standard courses. Over half of the course's content was standard, but it also presented a high topic variability (nine out of thirteen content categories were present) with similar levels of emphasis on the complementary content.

<i>Subcategory</i>	<i>Freq.</i>	<i>Summary</i>
Standard only	3	Almost entirely focusses on standard topics.
Standard*	1	Almost only uses standard topics but not in all four standard categories.
Standard + Topic	4	Primarily standard but with emphasis on a clear non-standard topic.
Standard* + Topic	4	A Standard + Topic course which uses three categories of standard content instead of the usual four.
Standard + Medley	1	Mostly standard and also equally uses a wide range of topics.

Table 4.3: Summary of the subcategories of Standard courses.

4.1.2 Medley courses

Medley courses account for 33% (7/21) of the identified ITP courses, with three subcategories arising: Medley Only (2/7), Medley + Standard (4/7), and Medley + Topic (1/7). A summary of these subcategories is given at the end of this subsection in 4.5. All seven of these courses presented over 60% variability. Six Medley courses had a standard deviation in the range of 3.95 to 7.78, with the exception of a particular course with a standard deviation of 9.11. This course's high standard deviation comes from the presence of two outlier content categories which account for 4% of the content used each, while most of the other categories contribute to around 10% of the content. Nevertheless, we deemed the course to have high enough variability (62%), and no clear emphasis on a particular content category, so it was categorised as "Medley".

Almost all of the medley courses (6/7) included the four standard content categories in their syllabi, with the exception of one course which did not include Logic. The three most common non-standard content categories present in these courses were Number Theory, Real Analysis, and Number Systems (6/7), followed by Complex Numbers and Group Theory (5/7). Table 4.4 shows the frequency of each content category and its average weight across the Medley courses.

<i>Content category</i>	<i>Frequency</i>	<i>Percentage</i>	<i>Average weight</i>
Proof Techniques	7	100%	10%
Functions and Relations	7	100%	9%
Set Theory	7	100%	5%
Logic	6	86%	9%
Real Analysis	6	86%	17%
Number Theory	6	86%	17%
Number Systems	6	86%	11%
Complex Numbers	5	71%	13%
Group Theory	5	71%	13%
Other	4	54%	3%
Linear Algebra	3	43%	10%
Discrete	2	29%	8%
Calculus	1	14%	17%

Table 4.4: Frequency and average weight of each content category, and the percentage of Medley courses that they appear in.

We now discuss the logistical information regarding Medley courses. These courses ranged between a weighting of 10 and 40 credits, with the single 40 credit course being delivered over the full academic year. Regarding assessment information, five courses split the students' final mark between coursework and an exam, and one relied purely on an exam (no information was found for one course). All of the courses for which we had information on student recommended input hours (6/7) kept to the standard of 10 hours per credit. Three courses provided information on textbook and authorship used, with these courses using one, two, and three textbooks respectively, with no commonalities between them. As for the reading lists, only one course of the seven provided information, with twelve books detailed on this list.

Medley Only

Medley Only courses accounted for 29% (2/7) of the Medley-labelled courses. In particular, one course presented all 13 different content categories, with a standard deviation of 3.93. This corresponded to a year-long course which aimed to impart a foundational knowledge of mathematics on students. The other (from our own institution) had an emphasis on Real Analysis and standard content (29% and 25% respectively), but also included topics in Number Theory, Number Systems, Complex Numbers and Discrete Mathematics. In context, this course is the first instance of pure mathematics

students encounter, and it aims to give students an introduction to abstract mathematics. Although there is a slight focus on two content categories, there is also a clear emphasis on a diverse syllabus, hence motivating us to label it as "Medley Only".

Medley + Standard

Medley + Standard courses account for 57% (4/7) of the Medley-labelled courses. In all of these, the weight of standard content was between 30% and 45%. Although there was emphasis on these topics, non-standard content constituted the majority of these courses.

Medley + Topic

One course was categorised as Medley + Topic. The Topic element was Number Theory, which accounted for 30% of the course content. This was the outlier course with a standard deviation of 9.11.

<i>Subcategory</i>	<i>Freq.</i>	<i>Summary</i>
Medley only	2	Presents a variability of 60% or higher with similar emphasis given to every topic used.
Medley + Standard	4	Has an high variability whilst using all four standard content categories
Medley + Topic	1	Like Medley + Standard but instead emphasises a non-standard topic

Table 4.5: Summary of the subcategories of Medley courses.

4.1.3 Topic courses

Topic-based ITP courses accounted for 19% (4/21) of the identified ITP courses. We subcategorised two as Topic Only courses and two as Topic + Topic courses. The main content categories used in these courses were Real Analysis, Number Theory, Number Systems, and Linear Algebra. Table 4.6 shows the frequency and average weight of each content category used in Topic-based courses, with the main content categories excluded. These categories appear maximally as main content in some courses and minimally in others. For example, Real Analysis appears once with 60% weight, and once with 20% weight giving an average weight of 40%, which is misrepresentative of its contribution across the four courses.

We made notes on the logistics of Topic courses. Topic courses ranged between 15 and 60 credits out of 120 credits. Three of the courses were worth 15 credits with a single outlier of 60 credits. This course was a full year course. Topic courses all used coursework, with exams making up the majority of the final grade. All courses adhered to the rubric of 10 hours of recommended student input per credit. In terms of the textbook and authorship used, no information was provided by any of the courses in this category. Further, no reading lists were provided for any of these courses.

<i>Content category</i>	<i>Frequency</i>	<i>Percentage</i>	<i>Average weight</i>
Logic	3	75%	13%
Functions and Relations	2	50%	6%
Proof Techniques	2	50%	14%
Calculus	2	50%	19%
Discrete	2	50%	5%
Set Theory	1	25%	7%
Other	1	50%	4%
Complex Numbers	1	25%	31%
Group Theory	0		

Table 4.6: Frequency and average weight of each content category, and the percentage of Topic courses that they appear in.

Topic Only

Topic Only courses make up half (2/4) of the Topic category. We identified a Real Analysis course, in which this category corresponded to 60% of the course content, with the second heaviest content category being Calculus (20% of the course content). This module included Logic, focusing on the use of quantifiers and logical operators, perhaps due to the common “epsilon-delta” proofs.. We also identified a Number Theory course, in which this category accounted for 63% of the course content. This course also included Logic and Proof Techniques. Neither of these courses used all of the standard content.

Topic + Topic

The other 50% (2/4) of the Topic category consists of Topic + Topic courses. We identified a Linear Algebra + Real Analysis course (26% and 22% of the course content respectively) and a Number Systems + Number Theory course. The latter’s syllabus mentioned a clear emphasis in number systems, but with no detail on the granular topics within Complex Numbers. Our computations revealed a content distribution of 31% Complex Numbers, 23% Number Systems, and 23% Number Theory. A more comprehensive look at the course’s syllabus led us to believe that Complex Numbers did not contribute significantly to the course’s content. Hence, in this case we chose to view Complex Numbers as an extension of Number Systems, providing the aforementioned categorisation.

<i>Subcategory</i>	<i>Freq.</i>	<i>Summary</i>
Topic Only	2	Predominantly uses topics from a non-standard content category.
Topic + Topic	2	Consists mainly of topics in two non-standard content categories.

Table 4.7: Summary of the subcategories of Topic courses.

<i>Course category</i>		<i>Description</i>	<i>Subcategories</i>
Standard 10	Standard Only 4	A mixture of logic, proof techniques, functions, relations, and set theory.	Standard 3 Standard* 1
	Standard + Topic 5	Standard content with a relevant emphasis on a particular content category.	+ Real analysis 1 + Number theory 3 *+ Number systems 1
	Standard + Medley 1	Standard content complemented by a wide variety of topics.	
Medley 7	Medley Only 2	A varied mixture of topics with no particular emphasis on any of them.	
	Medley + Standard 4	A mixture of topics along with a relevant emphasis on standard content.	
	Medley + Topic 1	A mixture of topics along with a relevant emphasis on a non-standard topic.	+ Number Theory 1
Topic 4	Topic Only 2	A significant emphasis on one particular content category.	Real analysis 1 Number theory 1
	Topic + Topic 2	A relevant emphasis on two different content categories.	Linear algebra + Real analysis 1 Number systems + Number theory 1

Table 4.8: Summary of the categories and subcategories of ITP courses in the Russell Group.

4.1.4 Further findings

In this subsection we present further results regarding the categorisation. We describe the distribution of standard content, course variability, and the standard deviation of content weights for the identified ITP courses. We provide summary statistics of all of these for each ITP course category to highlight the stark differences between them.

Distribution of standard content

Every ITP course identified used standard topics to varying degrees. As seen in tables 4.9 and 4.10, the distribution of these topic weights is significantly spread out. In a third of the courses (7/21), standard content accounted for less than 30% of the course syllabus. In around 40% of the courses, (9/21), it accounted for in between 30% and 60% of the syllabus. In around a quarter of the courses (5/21), standard content accounted for over 70% of the course syllabus. Table 4.11 shows the cumulative frequency of standard content weights in 10% bands and provides a simplified look at the

distribution of standard content. The average weight of standard content is 45.8%, and the median is 38.5%: in over half of the courses (12/21), standard content accounts for less than 40% of the syllabus. We are interested in the distribution of standard content within non-topic ITP courses, as it motivates our description of Medley-style courses (subsection 4.1.2). Table 4.12 shows summary statistics of standard content weights in each ITP course category.

<i>ITP Category</i>	<i>Weight</i>	<i>All?</i>
Standard*	85.9%	Yes
Standard	83.3%	Yes
Standard	78.7%	Yes
Standard	74.2%	Yes
Standard	72.0%	Yes
Standard	59.1%	Yes
Standard*	54.8%	Yes
Standard	53.8%	Yes
Standard	51.0%	Yes
Medley	44.7%	Yes
Standard	38.5%	Yes
Medley	36.2%	Yes
Medley	35.0%	Yes
Medley	30.8%	Yes
Topic	29.3%	No
Topic	28.3%	No
Medley	26.8%	Yes
Medley	25.0%	Yes
Topic	23.1%	No
Medley	21.6%	No
Topic	10.0%	No

Table 4.9: Weight of standard content in ITP courses and whether they used all four standard categories.

<i>Statistic</i>	<i>Figure</i>
Minimum	10.0%
1st Quartile	28.3%
Median	38.5%
Mean	45.8%
3rd Quartile	59.1%
Maximum	85.9%

Table 4.10: Summary of table 4.9.

<i>Interval</i>	<i>Frequency</i>
10% - 20%	1
20% - 30%	6
30% - 40%	4
40% - 50%	1
50% - 60%	4
60% - 70%	0
70% - 80%	3
80% - 90%	2

Table 4.11: Cumulative frequencies of table 4.9. Intervals are exclusive of the left-endpoint except for 10 - 20%.

<i>Statistic</i>	<i>Standard</i>	<i>Medley</i>	<i>Topic</i>
Minimum	38.5%	21.6%	10.0%
1st Quartile	54.1%	25.9%	19.8%
Median	65.5%	30.8%	25.7%
Mean	65.1%	31.4%	22.7%
3rd Quartile	77.6%	35.6%	28.6%
Maximum	85.9%	44.7%	29.3%

Table 4.12: Summary statistics regarding the weight of standard content for each ITP course category.

Variability in ITP courses

Every ITP course we found had a variability of at least 20.0% (i.e they all used at least 4/13 different mathematical content areas in their syllabi). As seen in table 4.13, almost a quarter of the courses (5/21) had a variability of 30.0% or less, just under 50% of courses (10/21) had variability between 30% and 60%, and around a third of the courses (6/21) had a variability of 60.0% or higher, with one clear outlier using all 13 different content categories. Both the average and median variability of these courses are 46.2%, which corresponds to the use of 6 different mathematical content areas.

Regarding the distribution of the standard deviations of content weights. As seen in table 4.16, around a quarter (5/21) of the courses have a standard deviation smaller than 7.52, with the same number having one greater than 10.39. The average standard deviation in ITP courses is 9.31, and the median is 8.91. These summary statistics are shown in table 4.14. Two clear outliers are present, corresponding to two courses categorised as Topic Only, which have standard deviations of 16.62 and 16.13 respectively. Table 4.17 shows summary statistics of the variability and standard deviation for each ITP course category.

<i>Category</i>	<i>Variability</i>	σ
Medley	100.0%	3.93
Medley	70.0%	5.56
Medley	61.5%	9.11
Standard	60.0%	6.59
Medley	60.0%	7.78
Medley	60.0%	7.07
Topic	53.8%	9.14
Medley	50.0%	7.21
Standard	50.0%	8.12
Medley	50.0%	7.52
Standard*	46.2%	10.94
Topic	46.2%	11.08
Standard	40.0%	8.91
Standard*	30.8%	12.82
Topic	30.8%	16.13
Topic	30.8%	16.62
Standard	30.0%	9.53
Standard	30.0%	8.80
Standard	30.0%	9.51
Standard	20.0%	8.66
Standard	20.0%	10.39

Table 4.13: Variability and standard deviation of content weights in identified ITP courses, ordered by descending order of variability.

<i>Statistic</i>	<i>Variability</i>	σ
Minimum	20.0%	3.93
1st Quartile	30.8%	7.52
Median	46.2%	8.91
Mean	46.2%	9.31
3rd Quartile	60.0%	10.39
Maximum	100.0%	16.62

Table 4.14: Summary of Table 4.13.

<i>Variability</i>	<i>Freq.</i>
20% - 30%	5
30% - 40%	4
40% - 50%	5
50% - 60%	4
60% - 70%	2
70% - 80%	0
80% - 90%	0
90% - 100%	1

Table 4.15: Cumulative frequency of Table 4.13. Intervals are exclusive of the left-endpoint except 20-30%.

<i>ITP Category</i>	<i>Variability</i>	σ
Topic	30.8%	16.62
Topic	30.8%	16.13
Standard*	30.8%	12.82
Topic	46.2%	11.08
Standard*	46.2%	10.94
Standard	20.0%	10.39
Standard	30.0%	9.53
Standard	30.0%	9.51
Topic	53.8%	9.14
Medley	61.5%	9.11
Standard	40.0%	8.91
Standard	30.0%	8.80
Standard	20.0%	8.66
Standard	50.0%	8.12
Medley	60.0%	7.78
Medley	50.0%	7.52
Medley	50.0%	7.21
Medley	60.0%	7.07
Standard	60.0%	6.59
Medley	70.0%	5.56
Medley	100.0%	3.93

Table 4.16: Variability and standard deviation of content weights in identified ITP courses, ordered by descending standard deviation.

<i>Statistic</i>	<i>Standard</i>		<i>Medley</i>		<i>Topic</i>	
	<i>Variability</i>	σ	<i>Variability</i>	σ	<i>Variability</i>	σ
Minimum	20.0%	6.59	50.0%	3.93	30.8%	9.14
1st Quartile	30.0%	8.70	55.0%	6.32	30.8%	10.60
Median	30.4%	9.21	60.0%	7.21	38.5%	13.61
Mean	35.7%	9.43	64.5%	6.88	40.4%	13.24
3rd Quartile	44.6%	10.18	65.8%	7.65	48.1%	16.25
Max	60.0%	12.82	100.0%	9.11	53.8%	16.62

Table 4.17: Summary statistics regarding variability and standard deviation of content weights for each ITP course category.

4.2 Results regarding the survey of student attitudes

In this section, we present the results from the survey of student attitudes as described in section 3.2. We present attitudes from students who have not yet been exposed to an ITP course (4.2.1), students who were exposed to a Medley course (4.2.2), students whose institution did not offer an ITP course (4.2.3), and students who were exposed to a Standard course (4.2.4). In each of these subsections we present results regarding students' overall and explicit attitudes towards proof, views on proof, and perceptions of the roles of proof. For the cohorts of second-year students, we also present results regarding their attitudes towards proof-based courses at their institutions.

In Chapter 2, we noted that one of the key reasons students struggle with proof is that they are not exposed to proof sufficiently during their secondary education. As such, we present attitudes towards proof based on students' previous qualifications at the end of subsections 4.2.1 and 4.2.2³. We restricted our attention to A-levels, SQA (Scottish Qualification Authority) Advanced Highers, and the International Baccalaureate (IB), as the sample sizes for other qualifications were insignificant.

4.2.1 Attitudes pre-exposure to an ITP course

Attitude towards proof

We first address the three Likert scale statements from question 7 regarding overall attitudes towards proof. Of the 64 responses we received from students who had not yet been exposed to an ITP course, the majority expressed confidence in their answers to *"I feel comfortable reading and understanding mathematical proof"*. Responses to *"I feel comfortable using techniques that I have learned to construct mathematical proof"* showed more variation. Further, a positive inclination was observed in the responses to *"I enjoy communicating ideas using mathematical proof"*. The responses to these statements are visualized in Figure 4.2. Considering the overall attitude scores by the method described in section 3.2.5 we found that 57.8% of respondents had an overall positive attitude, 17.2% had an overall neutral attitude, and 25.0% had an overall negative attitude (table 4.19). The attitude score with the highest frequency was 3, which aligns with the majority of overall positive attitudes (table 4.18).

We also examined combinations of responses to the Likert scale statements. As shown in Table 4.28, we grouped positive, neutral, and negative responses to "I enjoy communicating ideas using mathematical proof" and analysed the frequency of combinations of the responses to the other two statements. We see that clusters of responses arise. For example, in the first table (corresponding to positive responses to the third statement), responses are grouped in the top left, which indicates positive responses to the other two statements. Similar corresponding clusters arise in the bottom right of the last table. This suggests that there are links between the attitudes towards communicating, validating, and constructing mathematical proof.

³Note that we do not present these results for the remaining two populations of students as dividing these already small samples would provide no useful results.

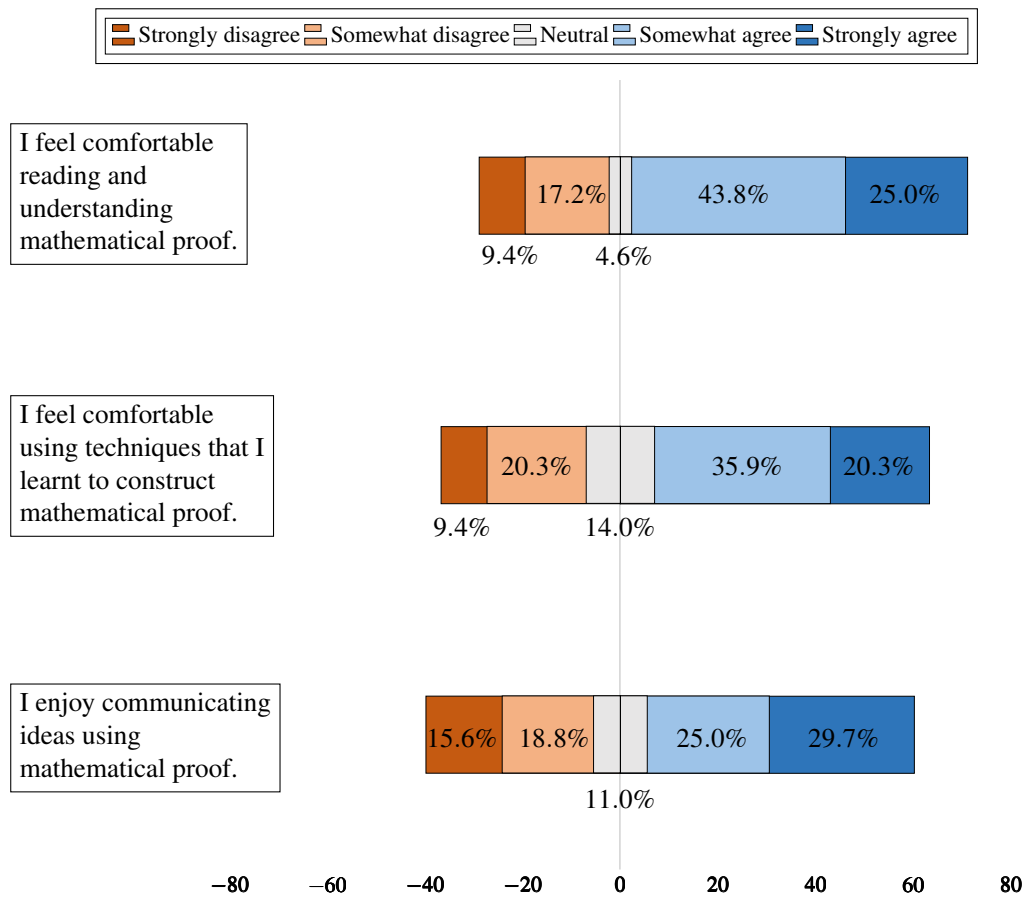


Figure 4.2: Responses to the Likert scale statements from students who have not yet been exposed to an ITP course.

Score	Freq.	%
6	7	10.9%
5	3	4.7%
4	8	12.5%
3	10	15.6%
2	9	14.1%
1	6	9.4%
0	2	3.1%
-1	3	4.7%
-2	5	7.8%
-3	3	4.7%
-4	3	4.7%
-5	2	3.1%
-6	3	4.7%

Table 4.18: Frequency of attitude scores from students who have not yet been exposed to an ITP course.

Overall attitude	Freq.	%
Positive	37	57.8%
Negative	16	25.0%
Neutral	11	17.2%

Table 4.19: Overall attitudes towards proof.

Explicit attitude	Freq.	%
Positive	5	7.8%
Negative	7	10.9%
N/A	52	81.3%

Table 4.20: Explicit attitudes towards proof.

What is a mathematical proof to you?

We now examine the open text responses to question 8, "*What is a mathematical proof to you?*". The explicit attitudes conveyed by the responses were dominated (81.3%) by neutral views, with 10.9% expressing a negative attitude and 7.8% a positive attitude (table 4.20). When analyzing students' views on proof (table 4.21), a significant 54.7% adhere to a formalist view, while only 10.9% subscribe to a subjective view. The remaining 34.4% of responses could not be classified and hence were categorised as *n/a*. The roles attributed to proofs align with the predominantly formalist view, with explanation (24.1%) and verification (17.2%) being the most cited (table 4.22 emphasises the correlation between these roles and students' views on proof).

<i>View on proof</i>	<i>Frequency</i>	<i>%</i>
Formalist	35	54.7%
Subjective	7	10.9%
N/A	22	34.4%

Table 4.21: View on proof of students who have not yet been exposed to an ITP course.

<i>Role</i>	<i>Total</i>	<i>Formalist</i>	<i>Subjective</i>	<i>N/A</i>
Verification	15	14	1	0
Explanation	21	16	4	1
Discovery	6	3	3	0
Systematization	9	7	1	1
Intellectual challenge	6	2	4	0
Communication	10	6	3	1
N/A	10			

Table 4.22: Total number of times that students who have not yet been exposed to an ITP course allude to different roles of proof, and whether these responses displayed a formalist or subject view.

Attitudes and views towards proof based on previous education

We now divide this population based on the highest level of qualification attained. Of the 64 respondents, 31.6% (20/64) had completed A-levels, 43.8% (28/64) had completed advanced highs, 10.9% (7/64) had completed the IB, and 14.1% (9/64) had completed other qualifications. Considering responses to question 7, figure 4.3 presents the responses to the Likert scale statements regarding general attitude toward proof, and table 4.24 presents the overall general attitude. Considering responses to question 8, table 4.25 presents whether responses had an explicitly positive, negative or neutral attitude towards proof, table 4.26 presents respondents view on proof, and table 4.27 presents perceptions on the role of proof. We do not include the results regarding combinations of responses to question 7 (as in table 4.28) or whether certain

roles of proof are seen as formalist or subjective (as in table 4.22). Repeating them with the divided samples rendered them impractical to interpret.

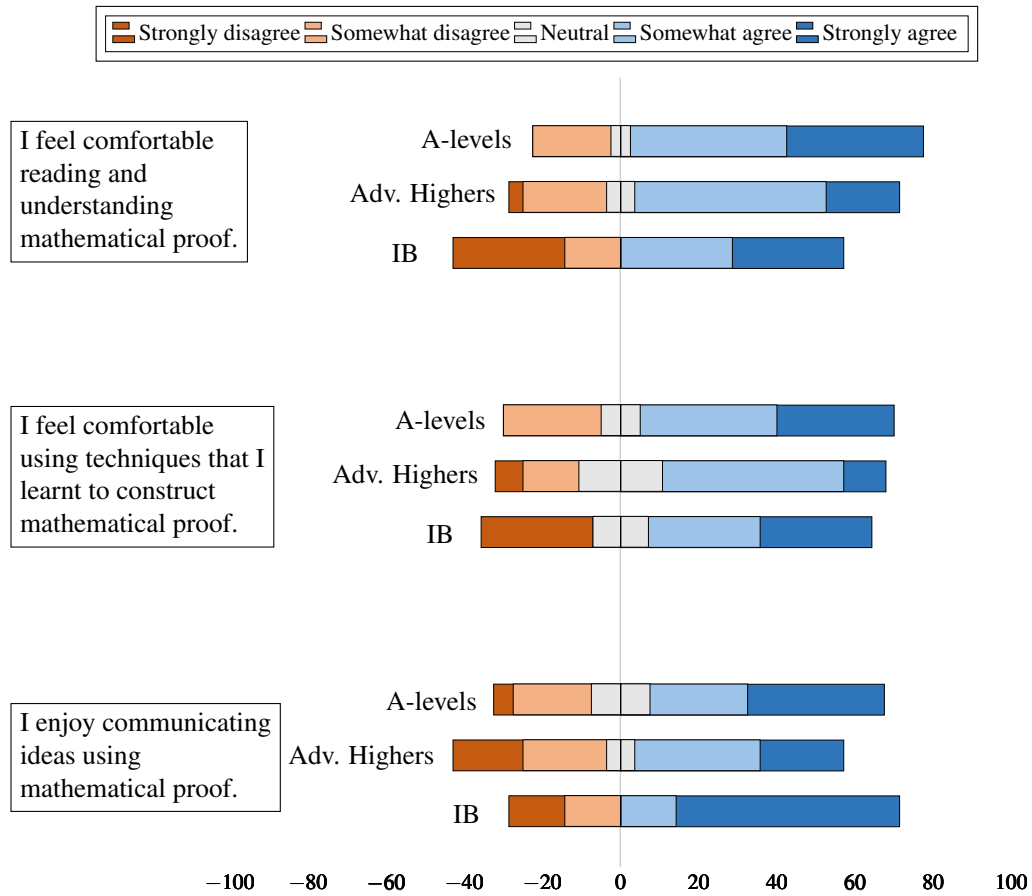


Figure 4.3: Responses to the Likert scale statements from students who have not yet been exposed to an ITP course based on previous education.

	<i>Previous education</i>	<i>Strongly disagree</i>	<i>Somewhat disagree</i>	<i>Neutral</i>	<i>Somewhat agree</i>	<i>Strongly agree</i>
<i>Statement 1</i>	<i>A-Level</i>	0.0%	20.0%	5.0%	40.0%	35.0%
	<i>A. Highers</i>	3.6%	21.4%	7.1%	50.0%	17.9%
	<i>IB</i>	28.6%	14.3%	0.0%	28.6%	28.6%
<i>Statement 2</i>	<i>A-Level</i>	0.0%	25.0%	10.0%	35.0%	30.0%
	<i>A. Highers</i>	7.1%	14.3%	21.4%	46.4%	10.7%
	<i>IB</i>	28.6%	0.0%	14.3%	28.6%	28.6%
<i>Statement 3</i>	<i>A-Level</i>	5.0%	20.0%	15.0%	25.0%	35.0%
	<i>A. Highers</i>	17.9%	21.4%	7.1%	32.1%	21.4%
	<i>IB</i>	14.3%	14.3%	0.0%	14.3%	57.1%

Statement 1: "I feel comfortable reading and understanding mathematical proof."

Statement 2: "I feel comfortable using techniques that I have learnt to construct mathematical proof."

Statement 3: "I enjoy communicating ideas using mathematical proof."

Table 4.23: Responses to the Likert scale statements of first-year students who have not yet been exposed to an ITP course, based on previous education.

<i>Overall attitude</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Positive</i>	70.0%	57.1%	57.1%
<i>Neutral</i>	10.0%	17.9%	14.3%
<i>Negative</i>	20.0%	25.0%	28.6%

Table 4.24: Overall attitudes from first-year students based on previous education.

<i>Explicit attitude</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Positive</i>	0.0%	14.3%	0.0%
<i>Negative</i>	10.0%	10.7%	14.3%
<i>N/A</i>	90.0%	75.0%	85.7%

Table 4.25: Explicit attitude towards proof of first-year students, based on previous education.

<i>View</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Formalist</i>	65.0%	50.0%	71.4%
<i>Subjective</i>	10.0%	7.1%	14.3%
<i>N/A</i>	25.0%	42.9%	14.3%

Table 4.26: Views on proof of first-year students, based on previous education.

<i>Role</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Verification</i>	25.0%	25.0%	20.0%
<i>Explanation</i>	35.0%	32.1%	10.0%
<i>Discovery</i>	10.0%	10.7%	10.0%
<i>Systematization</i>	5.0%	17.9%	10.0%
<i>Intellectual challenge</i>	5.0%	7.1%	20.0%
<i>Communication</i>	15.0%	7.1%	20.0%
<i>No role</i>	30.0%	42.9%	10.0%

Table 4.27: Percentage of responses that alluded to a role of proof, from first-year students based on previous education.

Strongly agree and somewhat agree					
	<i>Strongly agree</i>	<i>Somewhat agree</i>	<i>NAND</i>	<i>Somewhat disagree</i>	<i>Strongly disagree</i>
<i>Strongly agree</i>	10	2	0	0	0
<i>Somewhat agree</i>	2	10	4	2	0
<i>NAND</i>	0	3	0	0	0
<i>Somewhat disagree</i>	0	1	1	0	0
<i>Strongly disagree</i>	0	0	0	0	0

Neither agree nor disagree					
	<i>Strongly agree</i>	<i>Somewhat agree</i>	<i>NAND</i>	<i>Somewhat disagree</i>	<i>Strongly disagree</i>
<i>Strongly agree</i>	0	2	0	1	0
<i>Somewhat agree</i>	1	2	0	0	0
<i>NAND</i>	0	0	0	1	0
<i>Somewhat disagree</i>	0	0	0	1	0
<i>Strongly disagree</i>	0	0	0	0	0

Somewhat disagree and strongly disagree					
	<i>Strongly agree</i>	<i>Somewhat agree</i>	<i>NAND</i>	<i>Somewhat disagree</i>	<i>Strongly disagree</i>
<i>Strongly agree</i>	0	0	0	0	1
<i>Somewhat agree</i>	0	2	3	2	0
<i>NAND</i>	0	0	0	0	0
<i>Somewhat disagree</i>	0	1	1	6	0
<i>Strongly disagree</i>	0	0	0	1	5

Table 4.28: Frequency of each response combination of responses to the Likert scale statements from students who have not yet been exposed to an ITP course. Responses are grouped by positive, neutral, or negative answers to statement 3, "I enjoy communicating ideas using mathematical proof". Columns correspond to responses to statement 1, while rows correspond to responses to statement 2.

4.2.2 Attitudes post-exposure to an ITP course: Medley

Attitudes towards proof

We first address the three Likert scale statements from question [item 7](#) regarding overall attitudes towards proof. Out of the 76 responses we received from this cohort, the majority answered positively to *"I feel comfortable reading and understanding mathematical proof"*. Responses to *"I feel comfortable using techniques that I have learned to construct mathematical proof"* showed a similar positive majority. However, responses to the statement *"I enjoy communicating ideas using mathematical proof"* were more varied. The responses to these statements are visualised in [Figure 5.2](#). Considering the overall attitude scores, we found that 50.0% of respondents had an overall positive attitude, 23.7% had an overall neutral attitude, and 26.3% had an overall negative attitude. The most common attitude score was 4, which aligns with the majority of overall positive attitudes.

Score	Freq.	%
6	7	9.2%
5	4	5.3%
4	15	19.7%
3	9	11.8%
2	3	3.9%
1	8	10.5%
0	5	6.6%
-1	5	6.6%
-2	7	9.2%
-3	6	7.9%
-4	3	3.9%
-5	2	2.6%
-6	2	2.6%

Table 4.29: Frequency of attitude score and the percentage it represents with respect to the total number of responses from students who were exposed to a Medley course.

Overall attitude	Freq.	%
Positive	38	50.0%
Neutral	18	23.7%
Negative	20	26.3%

Table 4.30: Summary of [table 4.29](#).

Explicit attitude	Freq.	%
Positive	8	10.5%
Negative	15	19.7%
N/A	53	69.7%

Table 4.31: Explicit attitudes gathered from responses to [question 8](#) and the percentage it represents.

When examining combinations of responses to the Likert scale statements, as shown in [table 4.40](#), we make similar observations to those made in [subsection 4.2.1](#). This once again suggests that there are links between the attitudes towards communicating, validating, and constructing mathematical proof.

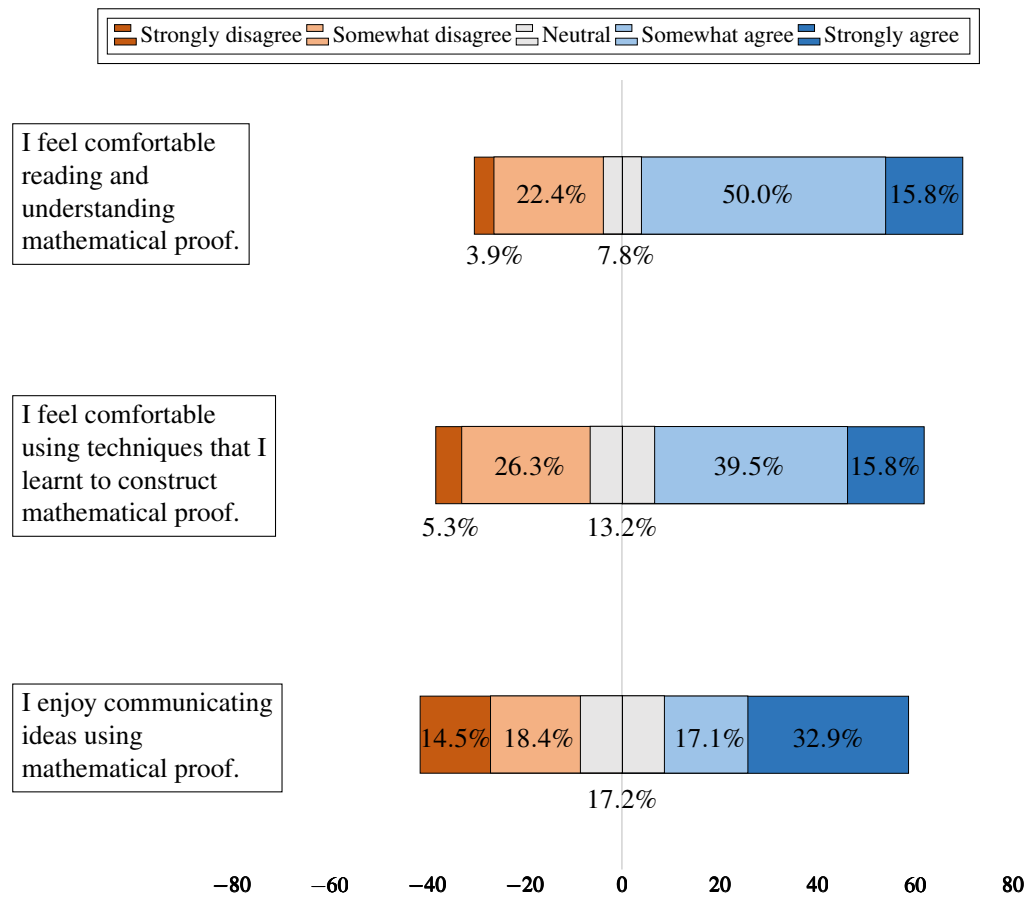


Figure 4.4: Responses to the Likert scale statements from students who have been exposed to a Medley course.

What is a mathematical proof to you?

We now examine the open text responses to question 8, "What is a mathematical proof to you?". The explicit attitudes conveyed by the responses were largely neutral (79.8%), with 10.5% expressing a positive attitude and 19.7% expressing a negative attitude (table 4.31). When analysing the students' views on proof (table 4.32), a considerable 40.8% of responses indicate a formalist view, with 28.9% exhibiting a subjective view and 30.3% being categorised as *n/a*. The roles of proof respondents most often cited, verification (27.4%) and explanation (24.5%), aligned with the predominantly formalist view (table 4.44).

<i>View on proof</i>	<i>Frequency</i>	<i>%</i>
Formalist	31	40.8%
Subjective	22	28.9%
N/A	23	30.3%

Table 4.32: Frequency of each view on proof from students who were exposed to a Medley course, based on attitude scores, and the percentage of responses in this population it represents.

<i>Role</i>	<i>Formalist</i>	<i>Subjective</i>	<i>N/A</i>	<i>Total</i>	<i>%</i>
Verification	25	4	0	29	23.4%
Explanation	12	14	0	26	32.8%
Discovery	3	0	0	3	9.4%
Systematization	5	2	1	8	14.1%
Intellectual challenge	0	1	1	2	9.4%
Communication	1	15	1	17	15.6%
N/A				21	27.6%

Table 4.33: The number of times that students allude to certain roles of proof in their open-text answers to question 8, and whether they possess more formalist or subjective views associated when they mention these roles.

Introduction to proof courses

71.1% of respondents answered *yes* to question 10, “*Is there a module that you feel particularly aimed to prepare you to read and write mathematical proof?*”, with all of them correctly identifying the ITP course they were exposed to. Considering question 11, the responses to “*This module succeeded in preparing me to read and write mathematical proof*” were: 24.1% strongly agree, 50.0% somewhat agree, 20.4% neither agree nor disagree, 1.9% somewhat disagree, and 3.7% strongly disagree (table 4.37), and the responses to “*I enjoyed this module*” were: 27.8% strongly agree, 25.9% somewhat agree, 14.8% neither agree nor disagree, 13.0% somewhat disagree and 18.5% strongly disagree.

<i>Response</i>	<i>Question 11a</i>		<i>Question 11b</i>	
	<i>Frequency</i>	<i>%</i>	<i>Frequency</i>	<i>%</i>
Strongly agree	13	24.1%	15	27.8%
Somewhat agree	27	50.0%	14	25.9%
Neutral	11	20.4%	8	14.8%
Somewhat disagree	1	1.9%	7	13.0%
Strongly disagree	2	3.7%	10	18.5%

Table 4.34: Frequency of each response to the Likert scale statements “This module succeeded in preparing me to read and write mathematical proof” (Question 11a) and “I enjoyed this module” (Question 11b), from respondents who perceived that they were exposed to a course which aimed to prepare them to read and write mathematical proof.

Attitudes and views towards proof based on previous education

We now divide this population based on the highest level of qualification attained. Of the 76 respondents, 56.6% (43/76) had completed A-levels, 28.9% (22/76) had completed Advanced Highers, 7.9% (6/76) had completed the IB, and 6.6% (5/76) had completed other qualifications.

Considering responses to question 7, Figure 4.5 presents the responses to the Likert scale statements regarding general attitude toward proof, and table 4.36 presents the overall general attitude. Considering responses to question 8, table 4.37 presents whether responses had an explicitly positive, negative or neutral attitude towards proof, table 4.38 presents respondents view on proof, and table 4.39 presents perceptions on the role of proof. We do not include the results regarding combinations of responses to question 7 (as in table 4.40) or whether certain roles of proof are seen as formalist or subjective (as in table 4.44). As in subsection 4.2.1, repeating these results with the divided samples rendered them impractical to interpret.

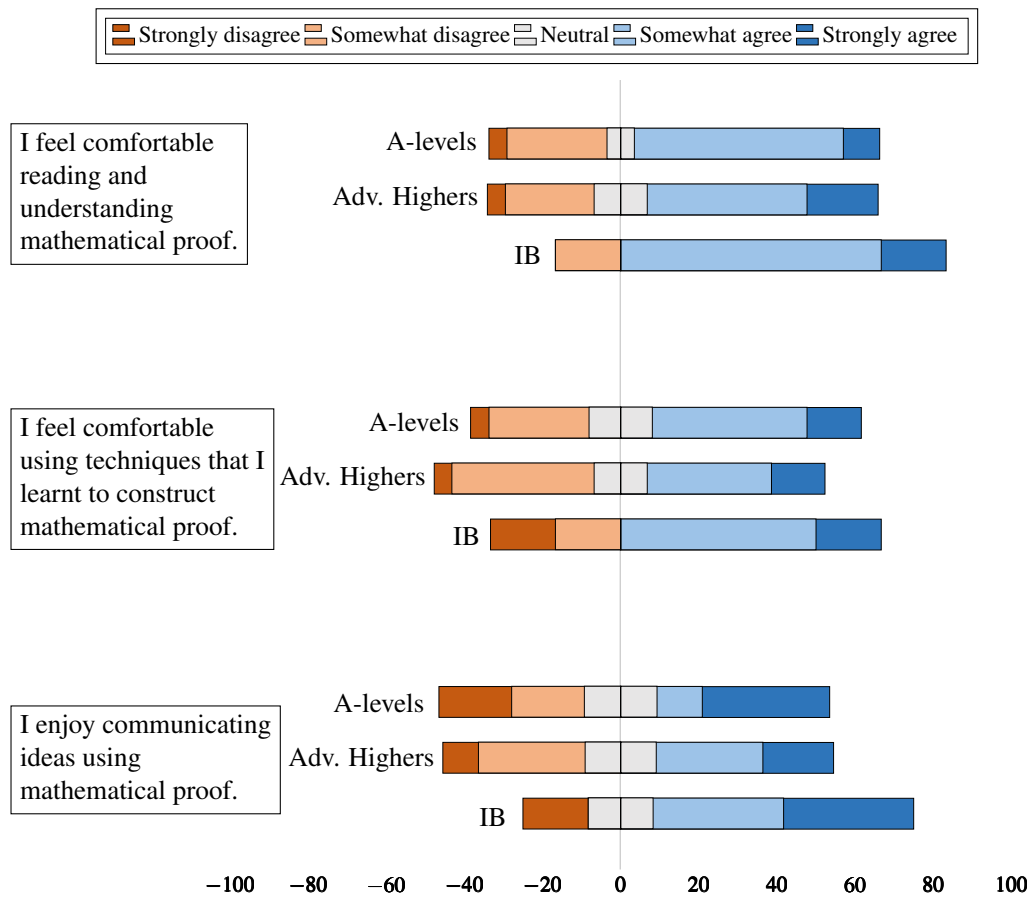


Figure 4.5: Y2 Attitudes per education level.

	Previous education	Strongly disagree	Somewhat disagree	Neutral	Somewhat agree	Strongly agree
Statement 1	<i>A-Level</i>	4.7%	25.6%	7.0%	53.5%	9.3%
	<i>A. Highers</i>	4.5%	22.7%	13.6%	40.9%	18.2%
	<i>IB</i>	0.0%	16.7%	0.0%	66.7%	16.7%
Statement 2	<i>A-Level</i>	4.7%	25.6%	16.3%	30.5%	14.0%
	<i>A. Highers</i>	4.5%	36.4%	13.6%	31.8%	13.6%
	<i>IB</i>	16.7%	16.7%	0.0%	50.0%	16.7%
Statement 3	<i>A-Level</i>	18.6%	18.6%	18.6%	11.6%	32.6%
	<i>A. Highers</i>	9.1%	27.3%	18.2%	27.3%	18.2%
	<i>IB</i>	16.7%	0.0%	16.7%	33.3%	33.3%

Statement 1: "I feel comfortable reading and understanding mathematical proof."

Statement 2: "I feel comfortable using techniques that I have learnt to construct mathematical proof."

Statement 3: "I enjoy communicating ideas using mathematical proof."

Table 4.35: Likert statement summary by previous education.

<i>Overall attitude</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Positive</i>	46.5%	40.9%	66.7%
<i>Neutral</i>	27.9%	22.7%	16.7%
<i>Negative</i>	25.6%	36.4%	16.7%

Table 4.36: View on proof by previous education level (Attitude scores) Y2.

<i>Explicit attitude</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Positive</i>	9.3%	18.2%	0.0%
<i>Negative</i>	23.3%	18.2%	0.0%
<i>N/A</i>	67.4%	63.6%	100%

Table 4.37: Explicit attitude towards proof by previous education level (Q8 text based responses) Y2.

<i>View</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Formalist</i>	39.5%	40.9%	33.3%
<i>Subjective</i>	32.6%	18.2%	50.0%
<i>N/A</i>	27.9%	40.9%	16.7%

Table 4.38: View on proof by previous education level Y2.

<i>Role</i>	<i>A-Level</i>	<i>SQA A. Higher</i>	<i>IB</i>
<i>Verification</i>	26.2%	30.8%	25.0%
<i>Explanation</i>	23.1%	19.2%	37.5%
<i>Discovery</i>	4.6%	0.0%	0.0%
<i>Systematization</i>	9.2%	3.8%	12.5%
<i>Intellectual challenge</i>	3.1%	0.0%	0.0%
<i>Communication</i>	18.5%	11.5%	12.5%
<i>No role</i>	15.4%	34.6%	12.5%

Table 4.39: Frequency of perceived role of proof by previous education level Y2.

Strongly agree and somewhat agree					
	<i>Strongly agree</i>	<i>Somewhat agree</i>	<i>NAND</i>	<i>Somewhat disagree</i>	<i>Strongly disagree</i>
<i>Strongly agree</i>	8	2	0	0	0
<i>Somewhat agree</i>	3	19	1	2	0
<i>NAND</i>	0	0	0	0	0
<i>Somewhat disagree</i>	0	2	1	0	0
<i>Strongly disagree</i>	0	0	0	0	0

Neither agree nor disagree					
	<i>Strongly agree</i>	<i>Somewhat agree</i>	<i>NAND</i>	<i>Somewhat disagree</i>	<i>Strongly disagree</i>
<i>Strongly agree</i>	0	2	0	0	0
<i>Somewhat agree</i>	0	2	2	2	0
<i>NAND</i>	0	1	0	1	0
<i>Somewhat disagree</i>	1	0	0	2	0
<i>Strongly disagree</i>	0	0	0	0	0

Somewhat disagree and strongly disagree					
	<i>Strongly agree</i>	<i>Somewhat agree</i>	<i>Neutral</i>	<i>Somewhat disagree</i>	<i>Strongly disagree</i>
<i>Strongly agree</i>	0	0	0	0	0
<i>Somewhat agree</i>	0	2	2	3	0
<i>Neutral</i>	0	0	1	3	0
<i>Somewhat disagree</i>	0	0	3	6	2
<i>Strongly disagree</i>	0	0	0	1	2

Table 4.40: Frequency of each response combination from students who were exposed to a Medley course. Responses are grouped by positive, neutral, or negative answers to “I enjoy communicating ideas using mathematical proof”. Columns correspond to answers to the question “I feel comfortable using techniques that I have learned to construct mathematical proof”, while rows correspond to answers to “I feel comfortable reading and understanding mathematical proof”.

4.2.3 Attitudes post-exposure to an ITP course: None

Attitudes towards proof

We first address the three Likert scale statements from question 7 regarding overall attitudes towards proof. Of the 13 respondents whose institution did not offer an ITP course, a distinct majority expressed confidence in their answer to *"I feel comfortable reading and writing mathematical proof"*. Responses to *"I feel comfortable using techniques that I have learned to construct mathematical proof"* were still largely confident, although this majority was less definitive. Further, a very positive response was seen to the statement *"I enjoy communicating ideas using mathematical proof"*. Considering the overall attitude scores for this cohort, we find that 84.6% of respondents had an overall positive attitude, 15.4% had an overall negative attitude, with no students showing an overall neutral attitude. The most common attitude score was 4, which aligns with the clear majority of overall positive attitudes. For this set of responses, we did not examine combinations of the Likert scale statements, as we had too few responses to determine any clear trends in this area.

A significant majority of students felt comfortable with reading and understanding proofs which suggests a foundational confidence in engaging with mathematical literature. In comparison, the data gathered from the second question reflects a potential gap between understanding and applying mathematical concepts, which could be addressed through targeted pedagogical interventions. Finally, the positive response towards the enjoyment of communicating ideas via mathematical proof, crucial for academic discourse in mathematics, implies that while students may not be formally trained in proof validation and construction, they value and take pleasure in the exchange of mathematical ideas.

What is a mathematical proof to you?

We now examine the open text responses to question 8, *"What is a mathematical proof to you?"*. The explicit attitudes conveyed were almost all neutral (92.3%), with 7.7% expressing a positive attitude and no responses expressing a negative attitude. When analysing students' views on proof (table 4.43), a considerable 46.2% indicate a formalist view, with 38.5% exhibiting a subjective view and the remaining 15.4% being characterised as *n/a*. The most cited roles of proof were verification (61.5%), system-

<i>Explicit attitude</i>	<i>Frequency</i>	<i>%</i>
Positive	1	7.7%
Negative	0	0%
N/A	12	92.3%

Table 4.41: Explicit attitudes gathered from responses to question 8 and the percentage it represents.

<i>Overall attitude</i>	<i>Total</i>	<i>%</i>
Positive	11	84.6%
Neutral	0	0%
Negative	2	15.4%

Table 4.42: Summary of overall attitudes for 4.2.3 cohort.

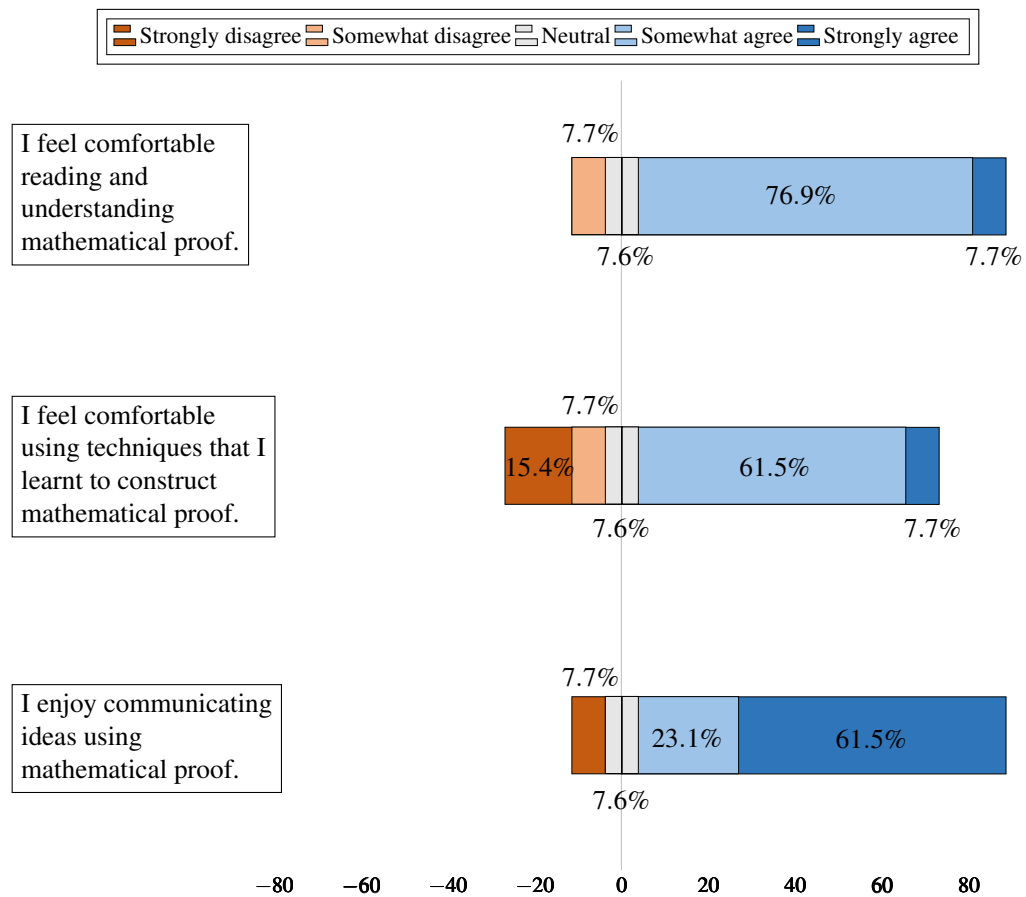


Figure 4.6: Y2 Birmingham Attitudes.

<i>View on proof</i>	<i>Frequency</i>	<i>%</i>
Formalist	6	46.2%
Subjective	5	38.5%
N/A	2	15.4%

Table 4.43: Frequency of each view on proof from students who were not exposed to an ITP course and the percentage of responses in this population it represents.

<i>Role</i>	<i>Formalist</i>	<i>Subjective</i>	<i>Total</i>	<i>%</i>
Verification	5	3	8	61.5%
Explanation	1	0	1	7.7%
Discovery	1	1	2	15.4%
Systematization	4	1	5	38.5%
Intellectual challenge	0	1	1	7.7%
Communication	0	5	5	38.5%
N/A			2	15.4%

Table 4.44: Frequency of each role of proof from students who were exposed to a medley course, and the percentage of responses in this population it represents.

atisation (38.5%) and communication (38.5%), in fitting with the relatively even split between formalist and subjective views.

Introduction to proof courses

Two students in this cohort responded *yes* to question 10 despite our characterisation of their institution as not delivering an ITP course. The students cited different courses in question 10. Both agreed to the Likert scale statement 11, "*This module succeeded in preparing me to read and write mathematical proof*", with one responding "strongly agree" and one "somewhat agree". Similarly, both students responded "strongly agree" to the statement item 11, "*I enjoyed this module*". This suggests that when students are able to connect with course material that explicitly addresses proof skills, they find it highly rewarding.

4.2.4 Attitudes post-exposure to an ITP course: Standard

We received one response from a student who was exposed to a Standard course (specifically Standard + Topic (Number Theory)). The overall attitude towards proof exhibited by the respondent was extremely positive, gaining the maximum attainable attitude score of 6. The response exhibited a subjective view on proof, and an emphasis on the roles of explanation and communication. This perception aligns with the proof-centered aims of the course's syllabus, of which two out of three are concerned with the communication aspect of mathematical proof (Figure 4.7). Moreover, this student correctly identified their institution's ITP course as a course that aimed to prepare them to read and write mathematical proof, and exhibited a neutral-positive attitude

towards it. Sadly, the lack of other respondents exposed to Standard courses renders this response unusable for any further analysis.

- Distinguish correct from incorrect and sloppy mathematical reasoning;
- Understand and write clear mathematical statements and proofs;
- To produce proofs using mathematical induction.

Figure 4.7: Aims which mention "proof", from the syllabus of the only Standard survey respondent.

Chapter 5

Discussion

The results presented in Chapter 4 raise questions about our Medley category of ITP courses. In section 5.1, we justify the necessity and existence of this category in light of the pre-existing categories, Standard and Topic. We go on to suggest that courses with these characteristics may cause students' attitude towards proof to become milder and promote them to adopt a content-independent view of proof in section 5.2. Then, in section 5.3 we discuss how a student's previous education may affect their attitude and perception of proof.

5.1 Justifying the Medley label

In this section, we address the labels used in our categorisation of ITP courses. In subsection 5.1.1, we examine standard content used to teach proof in the wider mathematical community, and see how our notion of standard content in ITP courses at Russell Group universities fits in. In subsection 5.1.2, we discuss why courses with Medley-style characteristics may be used to introduce students to proof, and critique their use in mathematics degree programmes at the Russell Group universities. Finally, in subsection 5.1.3 we justify the need of a Medley label by examining its relationship with the Topic label.

5.1.1 The notion of standard content

Here we examine the notion of standard content by discussing why topics in Logic, Functions and Relations, Proof Techniques, and Set Theory, seem to be favoured when introducing students to proof. Then, we examine this notion in the context of ITP courses at Russell Group universities, and the differences between ours and David and Zazkis [12]'s definitions of standard content.

As shown in sections 4.1.1 and 4.1.4, we found a distinct set of four mathematical content areas in ITP courses which were used in conjunction more frequently than any other content category. In fact, this notion of “standard” content in ITP courses is not new to the mathematical education community, as Selden [55] notes:

Mathematicians have written more than 25 transition-to-proof course textbooks: for example, Velleman [69] and Resek and Fendel [48]. Whilst their contents

vary, such courses and books often begin with a decontextualised treatment of logic emphasising truth tables and valid arguments, followed by a discussion of direct and indirect proofs, sets, equivalence relations, functions, and mathematical induction. After that, the books diverge, selecting specific mathematical areas, such as graph theory or number theory, in which students are to practise proving.

But why would content like this be preferential in the teaching of mathematical proof? Historical reasons may explain the use of Logic and Proof Techniques, in particular, the Western mathematical tradition founded by the ancient Greeks. Grabiner [23] suggests that Greek mathematicians developed both logical (axiomatic) proof and indirect styles of proving due to the cultural emphasis on logical thinking, the influence philosophy had in mathematics, and the seeking of ways to reduce problems to a simpler form [36]. The successors of Greek mathematics (medieval Islamic thinkers, and 17th and 18th century European intellectuals, for example) carried forward the idea of logical proof as the means to establish truth:

Once the Greeks had established this ideal of logical proof in mathematics, the ideal took on a life of its own. Even before Euclid, Aristotle was advocating the ideal of a science based on demonstration. Since mathematicians apparently had achieved truth by means of proof, practitioners of other areas of Western thought wanted to do the same. So thinkers, in theology, politics, philosophy, and science tried to imitate the mathematicians' method. [23]

These ideals for proof were then carried through the 19th and 20th century, and are now at the core of modern mathematics. Hence, it may not come as a surprise that the language of logic and different techniques of proof are perceived as essential knowledge for the proving process. Practical reasons may explain the use of Set Theory and Functions and Relations. These topics are an essential part of the abstract subjects of analysis and algebra [2], topics that are common in compulsory modules of undergraduate mathematics degrees. We often found that ITP courses not only served the purpose of introducing students to proof, but also acted as an introduction to abstract mathematics, so it makes sense to include these topics in an ITP course¹.

Our findings on the distribution of Logic, Proof Techniques, Set Theory, and Functions and Relations seem to corroborate that the notion of standard content is widespread amongst mathematical educators in the UK: Each identified ITP course used these categories in some way, with 66% (14/21) using all four. However, although the notion of standard content which we defined for Russell Group Universities has similarities to that of Selden [55], it differs from the one described in David and Zazkis [12], which defines standard content as the following topics:

“(...) symbolic/formal logic, truth tables, propositions, quantifiers, methods of proof (including contradiction and induction), number systems, sets relations and functions, infinite sets, and cardinality”.

Our definition of standard content did not include any topics from Number systems. We, the researchers, were introduced to proof in tertiary education through a Medley

¹Historically, analysis has been a subject at the core of philosophical debates regarding proof, in particular regarding changes to the *standards* of proof. This reinforces the idea that “proof takes place always in a historical, philosophical, cultural and social context” [23].

course at our own institution, so our notion as to what fundamental content should be used to teach proof was potentially impacted by this. However, our numerical findings support the fact that topics in Number Systems are not part of the definition of standard content in ITP courses in Russell Group universities. We encountered Number Systems in a medium number of courses (14/21) with small weight contributions towards course syllabi on average (12.9%). Moreover, we found that another content category, Real Analysis, appeared in the same number of ITP courses (14/21) with slightly higher average weights (18.4%), making it a stronger contender to be labelled as a standard category than Number Systems considering purely numerical-based findings. Due to these results, we had no systematic justification that we could use to include Number Systems as standard content without also including Real Analysis. Therefore, it was decided to include neither Number Systems nor Real Analysis in what we consider to be standard content.

At this point, we must also consider the limited sample size which the Russell Group Universities constitute as a limitation of our project. Selden [55] and David and Zazkis' [12] notion of standard concerns the wider mathematical community and a much larger set of high research output universities (176 identified ITP courses in R1 and R2 universities in the US), while our notion of Standard only concerns fraction of British universities. Therefore, as the courses that we identified at Russell Group universities may not provide an appropriate representation of all of the ITP courses at British universities, our conclusions regarding what denotes standard content should be treated with care. For example, it could feasibly be the case that our sample is more Analysis-heavy than the sample of ITP courses at non-Russell Group universities, maybe giving validity to the inclusion of Number Systems into our definition of standard content. Nevertheless, our findings could potentially provide an appropriate representation of these ITP courses amongst the current population of high-research driven universities (the British Russell Group, the American R1 and R2, etc.) due to similarities with the findings of David and Zazkis [12]. Both David and Zazkis' and our notions of standard content coincide extensively, and similar findings regarding Topic courses suggest that both processes of categorisation share similarly valuable aspects².

5.1.2 Necessity of the Medley label

We may question our definition of a Medley course, and whether it warrants its own ITP course label. Moreover, why would medley-style courses even be chosen to introduce students to proof? Are the characteristics we identified relevant enough to form their own category? Why has other literature not identified them before? In this section we aim to answer these questions, and justify the presence of the Medley category.

²Note that David and Zazkis' notion of standard content could be heavily influenced by their positions as highly experienced academics in the field. Indeed, whilst few would argue whether "Functions and Relations" is a fundamental part of mathematics as a discipline, a topic's designation as standard should be focused on the regularity of its appearance, rather than its hypothesised importance to mathematics. Questions and criticism regarding David and Zazkis [12]'s categorisation process are discussed further in sub-section 5.1.2.

Introduction to Proof courses aim to transmit broad sets of skills that are deemed beneficial for proving, including "sense making, learning proving techniques, precision, reading and validating proofs, and flexibility in the proving process" [53]. Meta-mathematical abilities³ have also been deemed as an important part of the proof-learning process. For example, Weber [71] examines the ability to handle mathematical concepts in an effective way within a mathematical proof. Further, Iannone [34] examines views of mathematicians regarding student approaches to proof production, and suggests that tertiary educators in the U.K. recognize "that different mathematical concepts call for different usages⁴ in proof production". The interviews analysed in the study⁵ regard a wide variety of topics and content areas in mathematics, reflecting that the meta-mathematical abilities that are required to be a successful prover are not contained within a single school of mathematical knowledge and could even be seen as content-independent. Savic's [53] interviews convey a similar but more content-specific finding: the mathematical community agrees that different content areas require different modes of reasoning and proving techniques, a point further accentuated by Dawkins and Karunaran [14].

While a mathematician's progress on a proof is believed to depend on their chosen approach to it [8], it is also the case that having an appropriate repertoire of examples and factual knowledge is considered to be beneficial in aiding the proving process [55]. All of this literature points towards the idealised notion of the "successful prover" as someone who:

1. Has a set of meta-mathematical abilities applicable to a wide range of content, or ideally, a set of abilities that are content-independent.
2. Has an appropriate set of skills within the specific content area in which the proof is situated.
3. Has a robust set of examples and a bank of factual knowledge regarding the particular topic.

These three characteristics suggest that students would benefit from a more holistic and topic-wide introduction to proof. In fact, conscribing to a single topic may not be an effective way to structure an ITP course. For example, logic has been put into question as a topic used to introduce abstract reasoning skills, as students often deem it to be "too abstract" [9, 15]. Having said that, a thorough development of basic symbolic/formal logic may have benefits on a pupil's learning when analysed under a social-constructivist lens⁶ [17], and has also been suggested as an effective resource when combined with material from other content areas [9].

³Meta-mathematical knowledge concerns the aspects and processes of mathematical learning which are not of mathematical nature. For example, the correctness of a solution, the elegance and style used, etc. [50].

⁴This refers to *concept usage*, "which refers to ways one operates with the concept in generating or using examples or doing proofs" [38].

⁵These interviews were originally conducted during a previous study [33].

⁶We direct the reader to Wray [76] for an introduction on social constructivist theories of learning.

Therefore, if having a wider range of content in an ITP course is seen as beneficial, and if an ITP course having a narrow syllabus is believed to be negative for students, then it would be unsurprising if courses with a wide range of content were sometimes used to introduce students to proof. In fact, trends in the variability of ITP courses (4.1.4) together with the results regarding the distribution of courses' content category weights (Tables 4.13 and 4.17) provide further evidence to suggest that Medley should exist as an ITP course label in its own right.

So, if there are arguments for the use of medley-style courses within the mathematical community, and if we are able to identify ITP courses with such characteristics, then why has this category not appeared in past literature? One explanation may be the difference in the way in which this study and that by David and Zazkis [12] define different mathematical content areas. For example, we chose to consider Linear Algebra and Group Theory as two separate content categories, while David and Zazkis consider them to be a part of the wider category of Algebra. This may fail to capture the differences between these two areas of mathematics in the classroom, as they could be used to teach different aspects of the proving process. On the one hand, topics in Linear Algebra may be approached in a more computational way, rendering it more suitable for teaching inductive proofs and writing standards (showing one's work step-by-step, clarity of results, etc.). On the other hand, group theory is seen as a more abstract subject within algebra, so it may appear in a course syllabus in order to teach abstract reasoning and argumentative techniques (for example, contradiction). Essentially, these two different areas, although they both belong to the wider category of Algebra, may have completely different uses in the classroom, and so proficiency in one may not guarantee proficiency in the other. Hence, a more granular look at the content used in ITP courses, where the pedagogical strategies used to deliver different content areas are considered, allows for better profiling of populations of university mathematics students.

Moreover, the different definitions of mathematical content areas proposed by David and Zazkis [12] may fail to correctly portray the nature of some ITP courses. In fact, part of the motivation for the early stages of this project was to find an appropriate label for the ITP course at our own institution, as we felt that David and Zazkis' methods failed to provide one. Their category that most effectively describes this course is "Sampler", which was not applicable to our sample of Russell Group ITP courses (4.1). We present the syllabus of this course, which we categorised as Medley, in Figure 5.1⁷.

5.1.3 Topic courses validating the Medley label

The literature supports the use of Topic courses to introduce students to proof. All the interviewees in Savic's report[53] state that one content-focused (Topic) course could satisfy the requirements of an introduction to proof course. Nevertheless, this view

⁷Although we criticize some aspects of David and Zazkis [12], it is worth noting that it is the only piece of literature that we found which attempts to categorise ITP courses. Given this, it is natural that discrepancies arise, which is by no means a discredit to David and Zazkis, but rather a contribution.

1. Sets, proof quantifiers, real numbers, rationals and irrationals.
2. Inequalities, mathematical induction.
3. Upper Bounds, Least Upper Bounds and Limits.
4. Monotone Convergence, decimals, series.
5. Complex numbers, roots of unity, polynomial equations, fundamental theorem of algebra.
6. Euclidean algorithm, prime factorization, prime numbers.
7. Congruence, primality testing.
8. Equivalence relations, functions,
9. Counting and choosing, binomial coefficients, more set theory.
10. Permutations.

Figure 5.1: Topic description of the ITP course in our institution

has been criticised as it raises questions about students' ability to transfer their proving skill set from one content area to another. The presence of topic-based courses in Russell Group Universities seems to be consistent with their presence in R1 and R2 US Universities. David and Zazkis [12] reported that topic courses accounted for 19% of the ITP courses they identified. We also found that 19% of the ITP courses in RG universities were topic-based (subsection 4.1.3). Under the assumption that the categorisation in David and Zazkis [12] is correct under our definition and process in categorising courses as "Topic", this could have some important implications. Firstly, under this assumption, the group of R1 and R2 universities in the US have a high level of similarity to the Russell Group, giving validity to comparisons between the two studies. Further, it could be inferred that there are similarities between the US and UK university systems in their attitude of the effectiveness of teaching proof through the lens of a specific topic. Given the minority share which topic courses constitute in the UK and US, it could be argued that the general view in mathematical education literature that topic courses are less effective than those who teach under more varied content is either corroborated by those in education, or that this literature has had an impact on the teaching of proof.

Our approach highlights the difficulties in making a quantitative description of an ITP course. As such, we cannot say with assurance that our results mirror that of David and Zazkis [12], as the similarity in numbers of topic courses could be attributed to a difference of definition of what constitutes a "Topic" ITP course. The ambiguity between these and simply a topic course in the wider undergraduate mathematics programme (which does not aim to introduce students to proof) means there is potential for different categorisations to arise. Another factor which could lead to a difference in results is the different definitions for content categories (as mentioned in subsection 5.1.2). The breadth of content categories proposed by David and Zazkis [12] suggests that ITP courses in the US R1 and R2 group have a variability smaller than those in the Russell Group (where courses with minimal variability were clear outliers), so there is a finer line for what constitutes a Topic course.

In light of this difference in the UK Russell Group and US R1 and R2 university populations, and given the addition of the “Medley” category, it is useful to ask whether “Topic” is still an appropriate course category. Arguably, this change in the framework gives more validity to the “Topic” label. Whilst “Medley”, is denoted quantitatively by high variability amongst content categories and low content weights, “Topic” is denoted by low variability and high content weight. So, from a numerical perspective, Medley and Topic are somewhat opposite. As Medley does not infringe on the Topic share of the ITP population, the two categories are suitably distinct, and appearances of the quantitative characteristics of each in the Russell Group show that both are necessary in describing this population.

5.2 The impact of Medley courses on attitudes towards proof

In this section, we discuss inferences regarding student attitudes in light of the results presented in section 4.2, their validity, and in particular the effect of the content studied in an ITP module on the attitudes of the students who were exposed to them. As briefly outlined in 4.2.3, we will not discuss results regarding attitudes towards proof from students whose institution does not offer an ITP course. Instead, we have included a short discussion of them throughout 4.2.3. Similarly, we will not address the results from students exposed to a standard course (4.2.4) since we only received one response from that population.

Prior to any specific analysis or discussion however, we make points on the limitations of the report in relation to the survey which are not pertinent to a specific cohort of students. Firstly, we note the high volume of responses to “*What is a mathematical proof to you?*” with a lack of positive or negative attitude (classified as n/a in [Table 4.31](#)). It can be hypothesised that some students interpreted the question as desiring a formal definition of mathematical proof, rather than their opinion on it (the desired interpretation). This would lead to responses largely lacking in emotive descriptors, instead containing more objective statements as we observed. Perhaps a reformulation of the question as “*How do you feel about mathematical proof?*” would encourage respondents to share more emotive responses in the desired manner. However, the use of both questions would most likely give the most useful perspective, allowing us to see true reflections of student attitudes towards proof while also examining their views on proof and perspectives of the role of proof. Another limitation is the fact that we refrained from conducting qualitative analysis on the open text responses. While we perceived common themes amongst the responses, the time and length constraints of this project did not allow us to explore this further.

5.2.1 Attitudes towards proof: Year 1 Edinburgh

Year 1 students at Edinburgh display a generally positive but varied engagement with mathematical proofs, with a significant inclination towards neutrality ([Figure 4.4](#)). The observed engagement towards mathematical proofs could be attributed to several fac-

tors. First, their courses focus a lot on the basics of mathematics, which might not leave much room for thinking about proofs in a deeper or more personal way. The way the curriculum is set up, students are more often shown how to do proofs step-by-step rather than encouraged to think about them creatively. Since these students are just starting out, they might not yet fully grasp the subtler aspects of proofs, which takes time and experience. The teaching methods used might not always show students how interesting or creative proofs can be, focusing more on the rules of how to do them. Also, moving up to tertiary mathematics education can be difficult, and students might stick to what feels more solid and straightforward at first. Moreover, if students are not exposed to the wider world of mathematics, including its history and different approaches, they might not see proofs as anything more than a way to show something is true, missing out on their broader significance. Changing how proofs are taught and making the curriculum more engaging could help students see the beauty in proofs and become more interested in them.

5.2.2 Attitudes towards proof: Year 2 Edinburgh

In light of our commentary on Medley courses in subsection 5.1.2, we now discuss the implications of this style of course in reference to the attitudes of students exposed to them. On average, Medley courses use around 9 different content categories to introduce students to proof (Table ??). Although these categories appear with different weights, the use of nine different mathematical areas may present a challenge to students, inadvertently imposing high cognitive loads which could impair learning and generate negative attitudes [75]. Further, on average, a British university semester lasts 11 weeks, which would require fast pacing of such courses, covering nearly one content category per week. Further, based on the logistical data found in 3.1.1, it can be seen that these courses tend to have two lectures per week. Thus, students would likely have only two hours (as well as workshop and tutorial hours) in which they are taught content from each category. These issues may be evidenced by around a quarter (26.3%) of the responses we received from students who had taken a Medley course having a negative attitude score. Despite this, half of the responses (50%) had a positive attitude score, which perhaps alludes to second-year students' exposure to a variety of other mathematics courses. In contrast to the overall attitudes, we found that 19.7% of respondents indicated an explicitly negative attitude, whilst only 10.5% indicated an explicitly positive attitude. This could be explained by the psychological effect known as *negativity bias* [51], in which negative opinions are found to be more vocalised than their positive counterparts.

5.2.3 Changes in attitudes due to exposure to a Medley course

The results in sections 4.2.1 and 4.2.2 suggest that exposure to a Medley course affects the attitudes and perception students have towards proof. Nevertheless, the nature of such changes could simply be due to the exposure to any ITP course regardless of the content used. Weber and Czoher [73] note that often, expert mathematicians expect students to develop a sophisticated conceptual understanding of proof which aligns with their own perception of proof:

Many mathematics educators⁸ maintain that the way that proof is practiced in mathematics classrooms should be informed by, and compatible with, the way that proof is practiced by the mathematical community [13, 26, 30, 60].

This links to the role of the teacher in the mathematics classroom. While different theories of learning characterise this role differently [21, 46, 76], there is a shared agreement that a teacher's beliefs will influence those of the student [18, 40]. The observed changes in the views on, and perception of the role of proof in our surveyed students support the claim that ITP courses do have an effect on students' conceptual understanding of proof.

We observe that there is a noticeable change in the attitudes towards proof after exposure to a Medley course, in that attitudes become milder. Whilst we see minimal changes in responses to the statements "*I enjoy communicating mathematical proof*", in the responses to "*I feel comfortable reading and understanding mathematical proof*" and "*I feel comfortable using techniques that I have learned to construct mathematical proof*", we see significant increases in "somewhat disagree", "somewhat agree" and "neither agree nor disagree" responses, and decreases in "strongly agree" and "strongly disagree" (these changes can be seen in Figure 5.2). This could be attributed to a number of reasons. Firstly, the exposure to proof in a formal manner could lead students to overcome some of their anxieties related to proof. This is shown by the decrease in the proportion of those who strongly disagree with the above statements. Secondly, the high proportion of first-year students who show positive attitudes toward proof could be attributed to them being at the stage of "unconscious incompetence"⁹, in that their limited experience in the field of mathematical proof leads them to have a false sense of proficiency. Gaining experience of proof could enlighten students as to what they do not understand. So, as students become aware of the importance of proof in mathematics, we would expect their attitudes to become more reserved. This is supported by the decrease in the percentage of those who strongly agree with the above statements.

However, these arguments may fail to explain why we see that explicitly negative attitudes nearly double, explicitly positive attitudes stay relatively consistent, and n/a answers drop significantly (as in comparison between Tables 4.20 and 4.31). As mentioned in subsection 5.2.2, the increase in negative attitudes could be attributed to cognitive overload, which is directly linked to the content used in the course. However, the broad variety of contexts in which proof is presented could lead to an increased awareness of the varied aspects of proof, evidenced by the decrease in n/a responses. Hence, this Medley course could be seen to effectively expose students to proof as it promotes its students gaining an attitude on the topic.

As mentioned in subsection 5.1.2, the characteristics of Medley courses (high variability and similar content category weights) are heavily influenced by the idea of proof being a content-independent practice. A student must look beyond the technicalities of the content of each proving practice (i.e. "epsilon and deltas") and approach proof in a more conceptual way. Promoting proof in a content-independent way may further

⁸As in *researchers in mathematics education*.

⁹As in the four stage competence model proposed in Curtiss and Warren [11].

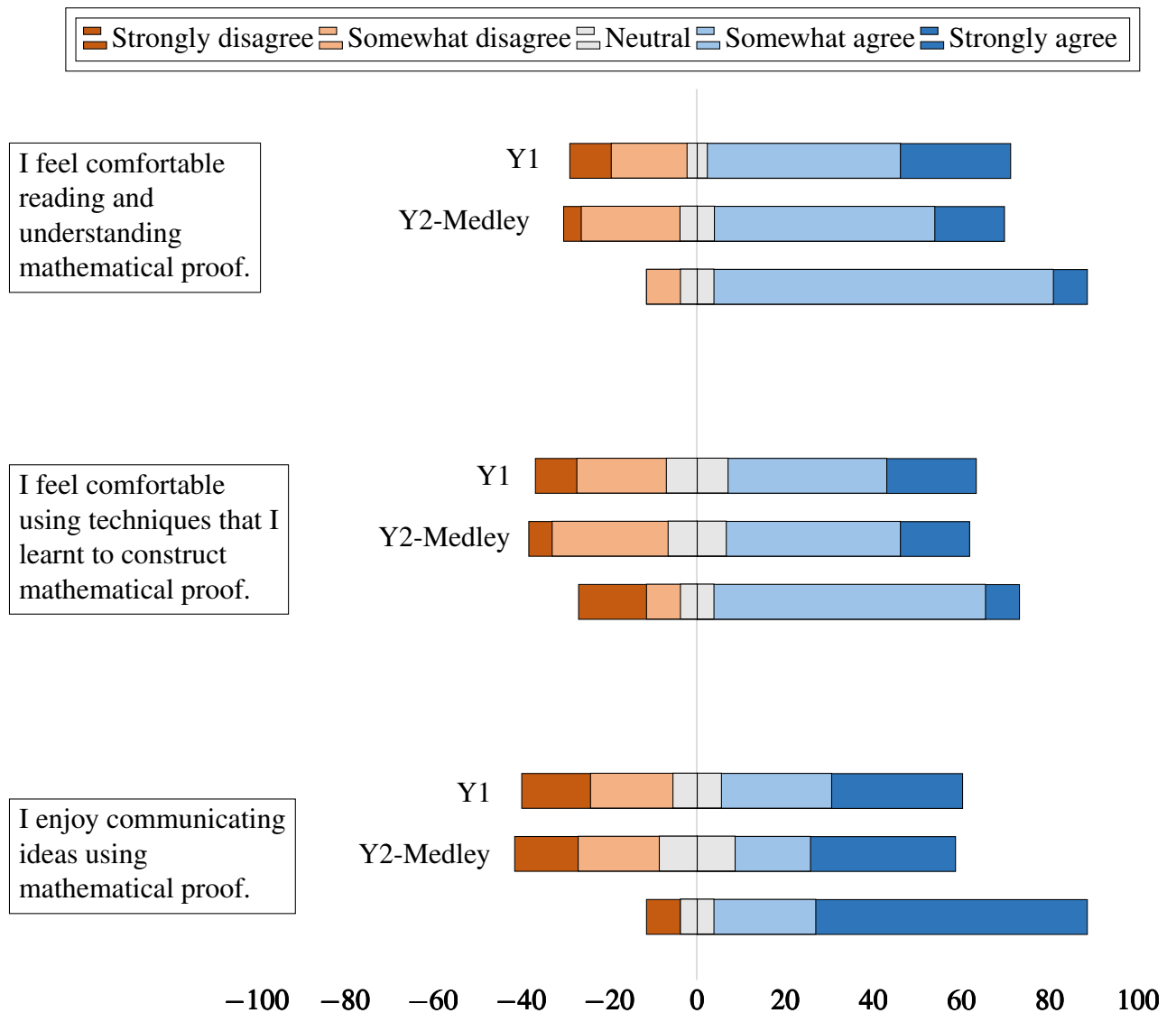


Figure 5.2: Y1 vs Y2 Edinburgh

emphasise the argumentation and conviction aspects of proof. The major increases in the perception of the roles of proof of verification (23.4% to 38.2%) and communication (15.6% to 27.6%) support this, but the lack of responses from other institutions prevents us from providing any definitive claims.

We observed a slight decrease in the number of responses with an explicit view on proof labelled as n/a (34.4% to 30.3%). Tentatively, this suggests that some students formed a view on proof, although we cannot determine what that view was. The decrease in the percentage of responses exhibiting a formalist view (54.7% to 40.8%) and the increase in subjective ones (10.9% to 28.9%) suggest that this Medley course promoted subjective views on proof. We would be able to better determine the impact that Medley courses have if we had been able to collect responses from other Medley courses.

5.3 Attitudes towards proof based on previous education

As shown in subsections 4.2.1 and 4.2.2 respectively, we gathered results based on previous education for students that have not yet been exposed to an ITP course, and those that were exposed to a Medley course. In this section, we restrict our attention to the first of these two populations, and discuss how a students' previous education may affect their attitude towards proof before they are exposed to an ITP course. In chapter 7, we discuss how we could have analysed the results from subsection 4.2.2.

5.3.1 Secondary education specifications

In this subsection we will consider the emphasis that the A-level, SQA Advanced Higher, and International Baccalaureate mathematics specifications put on proof¹⁰ in order to meaningfully interpret the results.

The A-level mathematics syllabus [67] only exposes students to some basic proof techniques and some algebraic proof in the context of trigonometry. Proof is mostly mentioned in the context of specific methods and examples, and there is also a light emphasis on the understanding of proof, for example, “students should be able to comprehend and critique proof”. A summary of how the A-level mathematics specification provided by the UK government mentions proof explicitly can be found in Appendix D.1.1.

It is worth mentioning that a potential failing of our survey was not including a separate option for A-level further maths, as further maths introduces proof by induction, which is presented in the contexts of sums of series, divisibility, and powers of matrices [66]. However, this is not an overly significant limitation when considering exposure to proof given that further maths emphasises methods rather than conceptual understanding. Nevertheless, it is important to consider that some respondents from this population might have spent a much larger part of their secondary education engaged with proof than others.

The SQA Advanced Higher syllabus [54] exposes students to some basic proof techniques, proofs related to differentiation and proof by induction. Rather than detailing the content that is covered in the course, this specification presents skills that students acquire whilst working towards the qualification. These skills mention methods and examples of proof that students should be familiar with. It is also worth mentioning that the idea of ‘rigorous proof’ is mentioned explicitly more than once. A summary of the skills detailed in the Advanced Higher mathematics specification that mention proof explicitly can be found in Appendix D.1.2.

Students who undertake the IB must complete one of two modules, ‘analysis and approaches’ or ‘applications and interpretation’, at either standard level or higher level

¹⁰The specifications that we collected are all up to date, and have not been changed since the cohorts that we surveyed undertook them.

[63]. Since we are only discussing responses from students at our own institution, we are able to assume that they all fulfilled the entry requirements detailed by the University of Edinburgh [64], and hence they must have taken ‘analysis and approaches’ at higher level. Therefore we will only consider the syllabus for that module. The syllabus for this module [42] exposes students to some basic proof techniques, proofs related to differentiation, proofs of geometrical properties using vectors, and proof by induction in a variety of contexts. Further, there is an entire section titled ‘Proof’ in the ‘Approaches to the teaching and learning of mathematics: analysis and approaches’ chapter that details the importance of including proof in the specification. Also, the document explicitly mentions that students should gain a ‘content-specific conceptual understanding that proof serves to validate mathematical formulae and the equivalence of identities’. A summary of how the content section of the Mathematics: analysis and approaches diploma programme specification guide provided by the International Baccalaureate Organisation mentions proof explicitly can be found in Appendix D.1.3.

Now comparing the amount that proof is explicitly mentioned in these syllabi, it could be concluded that the IB module exposes students to more proof techniques and examples than the other two, and that the Advanced Higher exposes students to more than the A-level does. It is also important to note the lengths of these documents when we consider the language used in reference to proof: the A-level specification is 27 pages long; the Advanced Higher specification is 47 pages long; and the Mathematics: analysis and approaches diploma programme guide is 100 pages long¹¹. It could be the case that the shorter specifications are written in a more concise manner, and therefore may emphasize the importance of the concept of proof less. Even so, regardless of whether we only consider the content detailed in the specification or the document as a whole, the IB module places the most emphasis on proof, followed by the SQA Advanced Higher, followed by the A-level.

5.3.2 The impact of previous education on attitudes towards proof

We now present some hypotheses as to what we might expect to observe when interpreting the results based on previous education from subsection 4.2.1, based on our comparison of the specifications. Then, we analyse these results to see if they align with our expectations.

Hypotheses based on mentions of proof in secondary education specifications

Firstly, we consider student attitudes towards proof based on their responses to the Likert scale statements in question 7. It is reasonable to assume that the amount of proof that a student is exposed to affects the amount that they cover in terms of both proficiency to understand and construct mathematical proof, and knowledge of methods and examples. Under this assumption, students have covered more if they undertook a qualification that places a heavier emphasis on proof. So, it may be the case that students who have covered more about proofs feel more comfortable working with them.

¹¹Unlike the A-level and Advanced Higher specifications, this document also includes an entire chapter dedicated to the teaching and learning of mathematics (although the Advanced Higher specification includes an Appendix detailing suggested teaching contexts for certain skills).

Therefore, we hypothesise that A-level students will have the most negative attitudes, while the IB cohort will have the most positive ones in response to the statements “*I feel comfortable reading and understanding mathematical proof*” and “*I feel comfortable using techniques that I have learned to construct mathematical proof*”. We have not made any predictions about different cohorts’ responses to “*I enjoy communicating mathematical proof*”, as proficiency to understand and construct mathematical proof may not imply an enjoyment of its communication.

We now hypothesise as to how exposure to these qualifications could affect students’ views on proof. The emphasis on methodology and specific examples in the A-level syllabus could lead to students gaining a formalist view of proof as it is presented with a focus on rules and exactitude. However, the mention of proof comprehension could lead some students to develop a subjective view, given the light emphasis that is placed on the necessity of understanding proof. The mentions of mathematical rigour in the Advanced Higher syllabus, both in terms of using a consistent process (formalist), and also emphasising clarity (subjective), lead us to expect a mixed set of views in this cohort. Finally, we expect the IB responses to be dominated by formalist views, as the focusses on the language and structure of proofs and the ability to validate statements resonate more with formalist ideals.

Finally, we consider how these qualifications may have an impact on what students deem to be the role of proof. We hypothesise that those in the A-level cohort will associate the role of proof with explanation due to the portrayal of proof as a method, and with communication since there is an emphasis on comprehending and analysing proofs. We expect those exposed to the Advanced Higher to perceive explanation as a key role of proof. Indeed, this syllabus places a heavy emphasis on “explaining and justifying concepts” [54]. The IB syllabus places importance on investigating and conjecturing statements, as well as validating mathematical formulae, and hence we would expect to see the roles of discovery and verification exhibited by this cohort.

Comparison of hypotheses and results

Given our hypotheses regarding attitudes towards proof, and given that thus far we have observed student attitudes over a wide range, the proportion of positive attitudes from students who completed A-Level is striking. Conversely, the IB students exhibit the most extreme negative attitudes, while Advanced Higher students have milder ones, as expected. Due to the University of Edinburgh’s demanding entry requirements, students who come to study mathematics at this institution must exhibit high proficiency in their previous qualification. Given that A-Level measures proficiency in proof as the competence in using proving methods, and as there is a positive relationship between achievement and attitude [62], it is understandable that these students demonstrate a positive attitude towards proof when they begin their tertiary education¹². The IB’s greater emphasis on proof may explain why we see a more representative breadth of attitudes (i.e both strongly negative and strongly positive). Moreover, these proportions may actually be due to the limited sample of respondents from this population (7

¹²This does not mean that their perception of proof aligns with the one held by the mathematical community.

students).

While our results do support the hypothesis of the IB syllabus promoting a formalist view on proof Table 4.26, the limitation regarding sample size must be considered, and hence we refrain from making any further comments about the cohort of IB students. Our hypothesis of A-Level students exhibiting a majority of formalist views, but also some subjective ones, is also supported by these results. The fact that half of the responses from SQA Advanced Higher students exhibit a formalist view supports our expectation that the views amongst this cohort would be mixed. However, a reason that we observe a high proportion of N/A responses (42.9%) could be that the subjective aspects of proof are communicated poorly to students because the wording of the specification may be interpreted differently by teachers across secondary schools¹³.

This idea also alludes to the high proportion of SQA Advanced Higher students not attributing a role to proof in their responses to question 8. Nevertheless, explanation was the most commonly mentioned role of proof as expected. The results from A-Level student also do not align with our hypothesis. In line with our hypothesis, we found explanation to be the role of proof most commonly alluded to. Nevertheless, verification accounted for 25.0% of the responses, communication only for 15.0% , and 30.0% were not attributed to any roles. This raises questions as to whether the aims of the A-Level specification are being communicated effectively to students.

¹³The SQA bases its syllabi and assessments on Scotland's Curriculum for Excellence (CfE), which has been intensively criticised in the previous years [35, 58] in part due to its ambiguous goals and wordings, and is currently undergoing a major reform. In fact, the SQA is being dismantled [74] following an OECD report reviewing Scotland's education policy [41, 43].

Chapter 6

Conclusion

The difficulty students experience when reading, constructing and communicating proof at the start of their tertiary education necessitate the requirement to introduce them to proof in a formal manner. Many of the Russell Group universities (21/24) do this via an ITP course, although the content used by these courses is distinctly varied from institution to institution. In answer to **RQ1**, we provide a categorisation of these ITP courses based on their content in Table 4.8, in which we find that:

1. The majority of ITP courses (9/21) restrict their content to topics in Functions and Relations, Logic, Proof techniques, and Set theory, content categories which we deem to be standard in introducing students to proof across the Russell Group.
2. A significant number of courses (7/21) choose to introduce students to proof through a wide variety of topics, with the aim of delivering a content-independent introduction to proof.
3. A small number of courses (4/21) choose to introduce students to proof in the context of a particular non-standard topic, for example, Real Analysis. Moreover, this finding is consistent with the ones in previous literature [12].

The most striking of these conclusions is that we found a distinct set of courses that did not fall under any of the labels provided by David and Zazkis' [12] categorisation, which motivated us to define a new ITP course label: "Medley".

Given these courses' nature as the first formal introduction students have to proof at university, it is natural to assume that they affect the attitudes of the students who take them. Further, in light of the need for content-specific research in mathematics education [14], we hypothesised that the content used in ITP courses may have an impact on the attitude students have towards proof. Although our conclusions are heavily limited in the fact that we do not discuss the impact that neither Standard nor Topic courses have on student attitudes, we answer **RQ2** by concluding that exposure to a Medley course may cause attitudes towards proof to become milder, and that these courses could succeed in promoting a content-independent view on proof.

Chapter 7

Future directions

Throughout this report, we detailed multiple limitations and restrictions that we encountered during our research, which in turn give rise to various avenues for further work in the field, some of which we detail in this chapter.

Avenues of analysis omitted

Statistical analysis of quantitative data

In section 4.1, we produced a number of summary statistics regarding the variability, average content category weights, and standard deviation of category weights, for each ITP course categorisation, and many statistics regarding attitudes on, views on, and perceptions of proof. Despite this, in this project we only conducted basic descriptive statistical analysis. The trends perceived in the data, like the distribution of standard content or the variability in ITP courses (Tables 4.13 and 4.9), leads us to believe that there are grounds to perform further statistical analysis on our results. For example, we could test whether the quantitative characteristics we attributed to each ITP course category are significant enough to justify the existence of our three ITP course labels (Tables ??, 4.17). To point towards future work, we have identified several advanced statistical analyses that could be pursued to deepen the understanding of our results:

1. Independent samples t-test to compare the differences in attitudes between two groups of students those taking different types of ITP courses to determine if there are significant differences in attitudes based on the ITP course type.
2. Analysis of variance (ANOVA), including both one-way and multifactor ANOVA, to assess whether different categories of ITP courses have varying effects on student attitudes and to consider the impact of multiple variables (e.g., course type, student background) simultaneously.
3. Correlation analysis using the pearson correlation coefficient to explore the relationships between student attitudes and other quantitative variables like course satisfaction or academic performance.
4. Regression analysis, such as linear regression for predicting student attitudes based on a set of predictor variables (e.g., course characteristics, student's edu-

cation backgrounds) and logistic regression for predicting binary outcomes, for example, whether students achieve a certain level of learning success.

Qualitative analysis of open text responses

Analyzing the open-text responses to question 8 using a qualitative method, instead of the deductive one we used (3.2.5) could have provided us with deeper insights into the personal views of these students that statistics alone cannot show. In fact, we perceived trends in the data portraying proof as a “tool” or in a “mechanical” way, which were not accurately captured by our deductive methods. Qualitative analysis would have allowed us to access subjective experiences, personal reflections, and specific feedback from students about their learning experiences. By carefully organizing and analyzing these responses, clear patterns and themes could emerge, which would show what aspects of ITP courses work well, and where there might be room for improvement. For example, students’ descriptions of which aspects of the course most significantly influenced their understanding of mathematical proof could indicate the effectiveness of various teaching methods. Similarly, feedback on the clarity, applicability, and engagement with course materials, like textbooks and supplementary resources, could provide direct insights as to how these aids support or hinder the learning process.

Relationship between aptitudes and attitudes towards proof

This project focused on the study of student attitudes. Given that mathematical educators have been exploring the relationship between students’ attitude towards and aptitude in mathematics and in particular, proof [20, 29, 37, 61], the results in our survey on student attitudes (4.2) could be used in the research of students’ aptitude in mathematical proof. As the mathematical education community is encouraging a content-specific approach to the discipline [14], our results constitute an appropriate contribution to the field. Not only did we produce a substantial set of data regarding the attitude students have towards proof in some Russell Group universities, but we did so under a content-specific lens, and hence our results point towards further work. Studies which may relate data regarding these student aptitudes, attitudes, and the type of ITP course they were exposed to could be of great importance. Conclusions surrounding what category of ITP course, or indeed what content covered in ITP courses is most effective at building students’ proficiency in proof could realistically be employed by university mathematics departments to improve their approach to teaching proof.

How a Medley course affects attitudes, depending on previous education

In section 5.3, we discussed how a student’s previous education might affect their attitudes towards, and perceptions of proof. Specifically, we analysed the cohort of students who had not yet been exposed to an ITP course, so that we can draw conclusions about how students feel about proof in the early stages of their transition to tertiary education. As we mentioned, we did not discuss the results presented in subsection 4.2.2 (attitudes of students who have been exposed to a Medley course based on previous education) since we found that the avenues to conduct this research were

too rich for the scope of this project. While this is the case, the clear continuation to our discussion of these results would be an analysis of the changes in attitudes before and after students are exposed to a Medley course based on their previous education. We would then explore whether the trends in the changes match those of the overall population. For example, we may ask the following: how does exposure to a Medley course affect the attitudes of students who undertook SQA Advanced Highers before attending university? Indeed, the results to perform this analysis already exist.

Final remarks

For clarity, this report serves as groundwork for many avenues of further research. ITP course categorisation can lead to more specificity in studies in this area, and questions arise from the many subsets of students created by this categorisation. A statistical analysis of the quantitative categorisation results is of particular interest, as is the relationship between the attitudes that we surveyed and students' aptitudes for proof, and also the impact that previous education has on their attitudes towards proof. Further, whilst the Russell Group describes a distinct subset of the population of UK universities which offer undergraduate mathematics degrees, it is feasible that universities outside of this subset whose focus is less cemented in research could have alternative approaches to introducing students to proof. Indeed, in two UK University Mathematics degree league tables in 2023 [10, 24], universities outside the Russell Group achieved the highest student satisfaction with teaching. As such, extending the ideas presented in this report, regarding ITP courses and their students, to the rest of the UK would provide a more representative perspective of introduction to proof in UK universities.

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Appendix A

Data Dictionary

Here we provide the data dictionary detailing how we determined what topics an ITP course covered based on what we observed in its syllabus. We note that in retrospect, the topic of fields could have been placed within Group Theory, and that other minor improvements could be made to our list of topics.

	<i>Topic</i>	<i>Mention in syllabus</i>
<i>Real analysis</i>	Bolzano-Weierstrass thm.	Bolzano Weierstrass Theorem
	Boundedness	Boundedness
	Continuity	Continuity, Continuity (also functions on subsets of the reals)
	Convergence and limits	Convergence, Convergence of Sequences and Series of Numbers
	Convergence tests	Cauchy criterion, series and convergence tests
	Exponential series	Exponential Series, Log and Power Functions
	Inequalities	Order Properties
	Infimum and supremum	Supremum, Upper Bounds
	Intermediate value theorem	Intermediate Value Theorem
	Limits of functions	Limits of Functions
	Monotonicity of sequences	Monotonicity
	Sequences	Sequence (real and complex), Sequences
	Series	Series
	Subsequences	Subsequences
<i>Calculus</i>	Differential equations	Differential Equations
	First principles	Calculus From First Principles
	Fund. thm. of calculus	Fundamental Theorem of Calculus
	Integrals	Integrals
	Mean value thm.	Mean Value Theorem
	Methods and optimization	Methods, Optimisation
	Multivariable calculus	Multiple Integrals, Multivariable Calculus
	Rolle's Thm.	Rolle's Theorem
	Standard derivatives	Standard Derivatives, The chain rule etc
	Vector derivatives	Derivatives (vector and scalar functions)

Complex number	Basic properties Complex conjugate De Moivre's thm. Exponential form Fund. Thm. of Algebra n^{th} roots Polar form Polynomial equations Roots of unity The complex plane	Basic Properties, Definition and Operations Complex Conjugate de Moivre's Theorem Exponential Form Fundamental Theorem of Algebra n^{th} roots Polar Form Polynomial Equations Roots of Unity The Complex Plane
Discrete	Binomial thm. Combinations Counting Permutations	Binomial theorem, binomial expansions, binomial coefficients Binomial coefficients, combinations, counting and choosing Counting Permutations
Functions and Relations	Bijections Composition Equivalence classes Equivalence relations Functions Injections and surjections Inverses Order relations Quotients Relations Restriction of functions	Bijections Composition Equivalence Classes Equivalence Relations Functions Injections, Injective Functions, Surjections, Surjective Functions Inverses Order Relations Quotients (simple examples in real life) Relations Restriction and Composition of Functions
Group theory	Applications Cosets Examples Fermat's little thm. Group homeomorphisms Isomorphisms Order Properties Quotient groups Subgroups	Applications Cosets Examples Fermat's Little Theorem Group Homomorphisms Isomorphisms Order Properties Quotient Groups Subgroups
Linear algebra	Basis and spanning sets Eigenvalues and eigenvectors Isomorphisms Linear maps Lines and planes in 3D Matrices Systems of linear equations Vector spaces Vectors	Bases, Linear Independence, Spanning Eigenvalues, Eigenvectors Isomorphisms Linear Maps Lines and Planes in 3-dimensional space Matrices Systems of Linear Equations Vector Spaces Vectors, Vector Spaces

<i>Logic</i>	Contrapositive	Contrapositive
	Converse	Converse
	Equivalence	Equivalence
	Implication	Implication
	Logical connectives	Logical Connectives
	Logical operations	Logical Operations
	Negations	Negation
	Quantifiers	Quantifiers
	Statements	Statements
	Truth tables	Truth Tables
<i>Number systems</i>	Algebraic and transcendental numbers	Algebraic and Transcendental Numbers
	Integers	Integers
	Irrational numbers	Irrational Numbers, Irrationality of $\sqrt{2}$
	Natural numbers	Natural Numbers
	Rational numbers	Rational Numbers, Definition of the Rationals
	Real number	Real Numbers
<i>Number theory</i>	Chinese remainder thm.	Chinese Remainder Theorem
	Congruence	Congruence
	Cryptography	Cryptography, Cryptography (DHM and RSA key exchange systems)
	Divisibility	Divisibility
	Euclid's algorithm	Euclid's Algorithm
	Euler's thm. and function	Euler's Function, Euler's Theorem, Fermat-Euler Theorem
	Fermat's last thm.	Fermat's Last Theorem
	Fermat's little thm.	Fermat's Little Theorem
	Fundamental thm. of arithmetic	Fund. theo. of arith. (unique prime factorisation) and applications, Fundamental Theorem of Arithmetic
	GCD and LCM	Greatest Common Divisor (GCD), Highest Common Factors
	Integer solutions of linear eqs	Integer Solutions of Linear Equations
	Modular arithmetic	Modular Arithmetic, Operations Modulo n
	Number of divisors	Number of Divisors
	Peano axiom	Peano Axiom
	Prime factorisation	Primality Testing, Prime Factorisation
	Prime numbers	Prime Numbers
	Pythagorean triples	Pythagorean Triples
	Units	Units
	Wilson's thm.	Wilson's Theorem
<i>Proof techniques</i>	Contradiction	Contradiction
	Contrapositive	Contrapositive
	Counterexample	Counterexample
	Direct proof	Direct Argument, Direct Proof
	Induction	Induction
	Proof structure	Proof Structure, Structure Analysis, Identifying Hypotheses, Conclusions

<i>Set theory</i>	Basic set theory	Algebra of Set Theory, Basic Set Theory
	Cardinality	Cardinality, Cardinality of a Set, Cardinality of the Power Set, Combinatorics
	Countability	Cantor's Diagonal Argument, Countability, Countability and Uncountability, Countable and Uncountable Sets
	Equality	Equality
	Principle of inclusion-exclusion	Principle of Inclusion/Exclusion
	Russell's paradox	Russell's Paradox
	Set operations	Set Operations
	Subsets	Subsets
	Union and intersection	Union, Union and Intersection, Venn Diagrams
	Well-ordering principle	The Well Ordering Principle

<i>Other</i>	Communicating mathematics	Communicating Mathematics
	Fields	Fields - Examples of Finite and Infinite
	Geometry	Geometry
	Polynomials	Polynomials - Multiplication and Long Division, Euclidean Algorithm for Polynomials, Remainder Theorem for Polynomials
	Simple axiomatic probability	Simple Axiomatic Probability
	Writing proofs in LEAN	Writing Proofs in Lean

Appendix B

Survey Consent and Data Protection

Here we detail the first four questions of the survey for completeness, and explain the reasoning behind each of them.

1. *Do you consent to us collecting and processing your personal data? [yes/no]*
If participants answer yes, then they are shown the next question. If they answer no, then the rest of the questions remain hidden, and clicking 'next' ends the survey.
2. *Are you a second-year student working towards a Mathematics degree? [yes/no]*
If participants answer yes, then they are shown the rest of the section. If they answer no, then the rest of the questions remain hidden, and clicking 'next' ends the survey.

*We will now ask you for two keywords to identify your data should you wish to withdraw it. **Do not include any personal details in your keywords.** If you want to withdraw your data, please email us with your two keywords.*

3. *What is your first keyword? [open text response]*
4. *What is your second keyword? [open text response]*

As stated in the preamble to these questions, questions 3 and 4 allow us to identify specific responses if we need to extract them without the participant unnecessarily imparting personal information.

Appendix C

Further Examples of Categorisation

C.1 Categorisation of a Standard only course

Figure C.1 shows the granular breakdown of the content categories used in the course and their weights. The first pie chart displays a moderate variety of content, with an emphasis on set theory. However, when we group standard content, we can see that this course places a heavy emphasis on it. This fact, along with the medium-low variability of 46%, led us to categorise this course as Standard only.

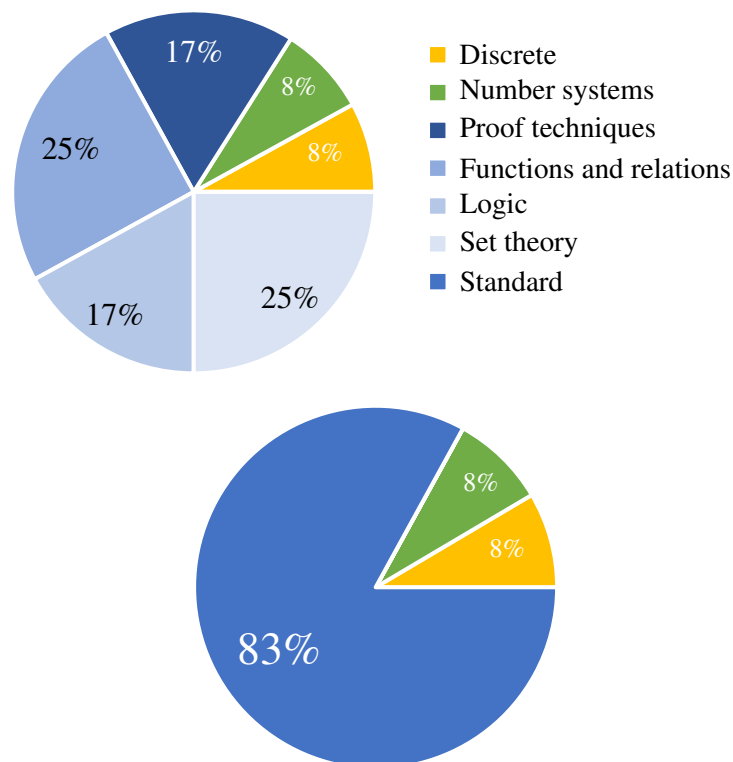


Figure C.1: Category weights of a Standard only ITP course first without, and then with standard content grouped.

C.2 Categorisation of a Topic + Topic course

Figure C.2 shows the granular breakdown of the content categories used in the course and their weights. This time we do not group standard content as only two standard content categories appear. The pie chart shows us that this course places a heavy emphasis on Complex numbers, and an equal emphasis on Number systems and Number theory. The medium-low variability of 46% suggests that we should not categorise this course as a Medley course, and because of the lack of standard content, we also should not categorise it as a Standard course. Therefore, this course must be a Topic course, but there is not enough emphasis placed on Complex numbers to label it as a Topic only course.

By taking a closer look at the syllabus for this course, we can see that, although complex numbers has a large weight, it is only introduced as another number system, so we can categorise this course as Topic + Topic, specifically Number systems + Number theory.

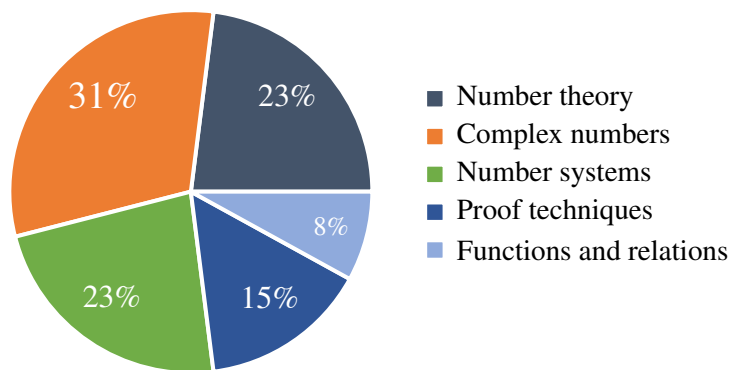


Figure C.2: Category weights of a Topic + Topic (Number Systems + Number Theory) ITP course.

Appendix D

Course Syllabi

D.1 Secondary education specifications

Here we detail any explicit mentions of proof in the secondary education specifications that we are considering.

D.1.1 Explicit mentions of proof in the A-level specification

A summary of how the A-level mathematics specification provided by the UK government [67] mentions proof explicitly:

- Students should be encouraged to construct mathematical proofs.
- Students should be able to comprehend and critique proofs.
- Students should understand and use the structure of mathematical proof and be able to use methods of proof including:
 - Proof by deduction.
 - Proof by exhaustion.
 - Disproof by counter-example.
 - Proof by contradiction.
- Students should be able to construct proofs involving trigonometric functions and identities.

D.1.2 Explicit mentions of proof in the SQA Advanced Higher specification

A summary of the skills detailed in the Advanced Higher mathematics specification provided by the Scottish Qualification Authority [54] that mention proof explicitly:

- Explaining and justifying concepts through the idea of rigorous proof.
- Disproving by counter-example
- Using indirect or direct proof in straightforward examples, specifically:
 - Proving a statement by contradiction.
 - Using proof by contra-positive.
 - Using direct proof.
- Using proof by induction

Two other occasions, that are not related to skills that students acquire during the course, on which proof is mentioned explicitly:

- An aim of the course is to clarify candidates thinking through the process of rigorous proof.
- In the context of calculus, students would benefit from exposure to formal proofs of differentiation (this point is only raised in an appendix that provides suggested learning and teaching contexts, so it is up to the teachers discretion as to whether they adhere to this suggestion).

D.1.3 Explicit mentions of proof in the International Baccalaureate specification

A summary of how the content section of the Mathematics: analysis and approaches diploma programme specification guide provided by the International Baccalaureate [42] mentions proof explicitly:

- Simple deductive proof, numerical and algebraic.
- How to lay out a left-hand side to right-hand side proof.
- Proof by induction in the context of complex numbers, differentiation, sums of sequences and divisibility.
- Proof by contradiction.
- Proof by counter-example
- Proofs of geometrical properties using vectors.

Other occasions, that are not in the content section, on which proof is mentioned explicitly:

- This course includes topics that are amenable to investigation, conjecture and proof.
- Content-specific conceptual understanding that proof serves to validate mathematical formulae and the equivalence of identities.
- Students will be introduced to the formal concept of proof.
- A section titled ‘Proof’ in the ‘Approaches to the teaching and learning of mathematics: analysis and approaches’ chapter. This section details the importance of proof and a summary of how it is mentioned in the syllabus.

Two occasions on which proof is mentioned as potential enrichment for students studying towards the International Baccalaureate:

- Proof of Pythagoras’ theorem in three dimensions.
- Proof of the cosine rule using the dot product.

D.2 Examples of ITP course syllabi

For the sake of interest, we include aims or descriptions of two ITP course syllabi.

D.2.1 Medley only

Description of a Medley only course that had the highest variability (100%) and smallest standard deviation of content weights (3.93) of all of the ITP courses:

This course offers a broad introduction to advanced mathematics covering topics from right across the subject, with a particular emphasis on mathematical communication and problem solving skills. It is an essential course for students on degree plans involving mathematics, physics or statistics.

D.2.2 Topic only

Aim of a Topic only course that had the highest standard deviation (16.62) of all the ITP courses:

The aim of this module is to introduce students to some of the basic ideas of number theory, and to use this as a content in which to discuss the development of mathematics through examples, conjectures, theorems, proofs and applications. The module will introduce and illustrate different methods of proof in the context of elementary number theory, and will apply some basic techniques of number theory to cryptography.

Aims of another Topic only course that had the second highest standard deviation (16.13):

- To begin the study of analysis, continued in MATH0004 and MATH0013. Mathematical analysis is one of the most important and well-developed strands of pure mathematics with many elegant and beautiful theorems, and also with applications to many areas of mathematics and mathematical physics.
- To introduce students to the ideas of *formal* definitions and *rigorous* proofs (one of the fundamental features of modern mathematics, and something that is not familiar from A-level), and to develop their powers of logical thinking.